

Enabling brain-wide mapping of layer-specific functional connectivity at 3T via layer-dependent fMRI with draining-vein suppression

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eLife Assessment

This **useful** study presents a possible solution for a significant problem - that of draining vein sensitivity in functional MRI, which complicates the interpretability of laminar-fMRI results. The addition of a low diffusion-weighted gradient is presented to remove the draining vein signal and obtain functional responses with higher spatial fidelity. However, the strength of the evidence is **incomplete**, and most tests appear to have been done only in a single subject. Significance thresholds in presented maps are very low and most cortical depth-dependent response profiles do not differ from baseline, even in the BOLD data shown as reference. Curiously, even BOLD group data fails to replicate the well-known pattern of draining towards the cortical surface.

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Abstract

Layer-dependent functional magnetic resonance imaging (fMRI) is a promising yet challenging approach for investigating layer-specific functional connectivity (FC). Achieving a brain-wide mapping of layer-specific FC requires several technical advancements, including sub-millimeter spatial resolution, sufficient temporal resolution, functional sensitivity, global brain coverage, and high spatial specificity. Although gradient echo (GE)-based echo planar imaging (EPI) is commonly used for rapid fMRI acquisition, it faces significant challenges due to the draining-vein contamination. In this study, we addressed these limitations by integrating velocity-nulling (VN) gradients into a GE-BOLD fMRI sequence to suppress vascular signals from the vessels with fast-flowing velocity. The extravascular contamination from pial veins was mitigated using a GE-EPI sequence at 3T rather than 7T, combined with phase regression methods. Additionally, we incorporated advanced techniques, including simultaneous multi-slice (SMS) acceleration and NOise Reduction with DIstribution Corrected principal component analysis (NORDIC PCA) denoising, to improve temporal resolution,

spatial coverage, and signal sensitivity. This resulted in a VN fMRI sequence with 0.9-mm isotropic spatial resolution, a repetition time (TR) of 4 seconds, and brain-wide coverage. The VN gradient strength was determined based on results from a button-pressing task. Using resting-state data, we validated layer-specific FC through seed-based analyses, identifying distinct connectivity patterns in the superficial and deep layers of the primary motor cortex (M1), with significant inter-layer differences. Further analyses with a seed in the primary sensory cortex (S1) demonstrated the reliability of the method. Brain-wide layer-dependent FC analyses yielded results consistent with prior literature, reinforcing the efficacy of VN fMRI in resolving layer-specific functional connectivity. Given the widespread availability of 3T scanners, this technical advancement has the potential for significant impact across multiple domains of neuroscience research.

Introduction

Layer functional magnetic resonance imaging (layer fMRI) is an emerging field which measures the layer-specific activity noninvasively in humans. The ability to extract layer-specific signal provides an exciting opportunity to dissociate between bottom-up feedforward (FF) and modulatory feedback (FB) responses which were activated in separated layers of a cortical unit. However, implementing layer-dependent fMRI is technically challenging. First, obtaining layer-specific detail requires much higher spatial resolution than standard fMRI, with voxel sizes typically in the sub-millimeter range. This smaller voxel size inherently reduces the signal-to-noise ratio (SNR), which is why most layer-dependent fMRI studies rely on 7T scanners for enhanced sensitivity (Brouwer et al., 2024 [↗](#); Dresbach et al., 2023 [↗](#); Lankinen et al., 2023 [↗](#)). Another challenge arises from the signal dependency across layers, particularly in blood oxygen level-dependent (BOLD) fMRI. BOLD contrast has been the gold standard for nearly three decades (Bandettini et al., 1992 [↗](#); Ogawa et al., 1990 [↗](#)) due to its ability to acquire rapid images using gradient echo (GE)-based echo planar imaging (EPI) (P. Mansfield, 1977 [↗](#)). While arteries, capillaries, tissues and draining veins all contribute to BOLD responses, the draining veins are considered the major contributor because of the larger volume and lower baseline oxygenation level compared to arteries (Boas et al., 2008 [↗](#)). This reliance on draining veins presents a significant obstacle for layer-dependent fMRI using BOLD contrast. Intracortical veins, which run perpendicular or parallel to the cortical surface, drain into pial veins. As a result, BOLD signals originating in the lower cortical layers can be carried into the upper layers, a phenomenon known as the ‘leakage model’ (Markuerkiaga et al., 2016 [↗](#)). Additionally, susceptibility changes near large pial vessels—often termed the “blooming effect”—can lead to extravascular BOLD signal distortions in voxels distant from the vessel (Kashyap et al., 2018 [↗](#); Li et al., 2012 [↗](#); Moerel et al., 2018 [↗](#)).

In contrast to GE-EPI, spin-echo EPI (SE-EPI) primarily detects signals from the microvasculature, as contributions from large veins are largely suppressed. This suppression improves spatial specificity and minimizes cross-layer contamination caused by large vessels (Boxerman et al., 1995 [↗](#); Duong et al., 2003 [↗](#)). However, SE-EPI generally exhibits lower functional sensitivity compared to GE-EPI because it refocuses BOLD dephasing, resulting in reduced overall signal changes during neural activation. Recently, a double spin-echo EPI method was introduced to enhance sensitivity in layer-dependent fMRI, achieving 0.8-mm isotropic resolution (Han et al., 2021 [↗](#)). This approach demonstrated cortical activation peaking at approximately 1.0 mm from the surface of the primary motor cortex, indicating better layer specificity compared to GE-EPI. However, the inherently longer echo time (TE) of SE-EPI leads to a longer repetition time (TR) compared to GE-EPI. For the double spin-echo EPI method, the slab thickness is 12.8 mm with an effective TR of 6 seconds, making it challenging for brain-wide studies. Additionally, the extended EPI readout time required for submillimeter-resolution fMRI introduces T2* contamination, partially diminishing spatial specificity.

To address the issues of draining-vein contamination and blooming effect, cerebral blood volume (CBV)-based approaches have been gaining popularity in recent years. These methods are based on the principle that CBV changes primarily originate from small arterioles located near neural activation sites within specific layers (Drew et al., 2011 [↗](#); Gagnon et al., 2015 [↗](#)). In contrast, larger downstream vessels are expected to have a limited contribution to overall CBV changes (Drew et al., 2011 [↗](#); Gagnon et al., 2015 [↗](#); O'Herron et al., 2016 [↗](#); Takano et al., 2006 [↗](#)). Recently proposed CBV-based approaches include vascular space occupancy (VASO) fMRI (Huber, Handwerker, et al., 2017 [↗](#); Lu et al., 2003 [↗](#), 2013 [↗](#)), integrated VASO and perfusion (VAPER) (Chai et al., 2020 [↗](#)), Magnetization transfer (MT) weighted laminar fMRI (Pfaffenrot & Koopmans, 2022 [↗](#)) and Arterial Blood Contrast (ABC) (Priovoulos et al., 2023 [↗](#)). The conventional VASO technique takes advantage of the difference between the longitudinal relaxation times (T1) of tissue and blood, measuring the remaining extravascular water signal at the blood-nulling time point along blood T1 recovery after an inversion pulse. Upon neural activation, an increase in CBV leads to a relative increase in the amount of nulled blood volume within the voxel, causing a negative signal change. VAPER technique employs DANTE (Delay Alternating with Nutation for Tailored Excitation) preparation module (Li et al., 2012 [↗](#)) to suppress fast-moving spins such as the vascular signal from arteries, veins but retain the stationary tissue signal. MT-weighted laminar fMRI enhanced the sensitivity to CBV by minimizing unwanted extravascular BOLD contributions from larger veins using MT preparation. Likewise, the ABC method enhances CBV weighting by selectively reducing venous and tissue signals through a pulsed saturation scheme.

Although CBV-based imaging approaches demonstrate high spatial specificity, the brain-wide acquisition time is relatively long due to the nature of the contrast-generation mechanism, which limits spatial coverage. Within the VASO fMRI framework, spatial coverage along slice direction is 18 mm and the acquisition time per slab is 3 seconds (Huber, Handwerker, et al., 2017 [↗](#)). Leveraging the findings that CBV weighting can be achieved without imaging at strict blood-nulling time (Ciris et al., 2014 [↗](#); Wu et al., 2008 [↗](#)), the MAGEC VASO technique (Huber et al., 2021a [↗](#)) further improved spatial coverage to 83.2 mm with 0.8 mm isotropic resolution, achieving a volumetric acquisition time of ~8 seconds, including both contrast preparation and signal readout. Other CBV-based approaches also require prolonged acquisition time for comparable spatial coverage. The VAPER method, for example, achieves an effective acquisition time of 12 seconds for 80.64 mm spatial coverage at 0.84 mm resolution (Chai et al., 2024 [↗](#)). MT-weighted layer fMRI requires 40.1 seconds for a 24-mm thick volume at 0.75 mm isotropic resolution, while the ABC method with 0.9 mm resolution covers a 27.9-mm thick volume in 2.7 seconds. In addition to CBV-based approaches, cerebral blood flow (CBF)-based methods, such as perfusion imaging with zoomed arterial spin labeling (ASL), have demonstrated layer specificity at 7T (Shao et al., 2021 [↗](#)). This method achieves a resolution of 1-mm isotropic with a slab thickness of 24 mm and an effective TR of 16.8 seconds. Among the CBV- and CBF-based techniques discussed above, the largest spatial coverage along superior-inferior axis is 83.2 mm. Achieving brain-wide mapping of layer-specific functional connectivity, however, requires a field-of-view (FOV) size along the superior-inferior axis (z-axis) of around 120 mm to cover the entire cerebrum (Mennes et al., 2014 [↗](#)). Assuming an inverse relationship between voxel size and acquisition time, we extrapolated acquisition times for the methods mentioned above based on a spatial coverage of 120 mm with a 0.9 mm isotropic resolution. Under these conditions, the volumetric acquisition times for MAGEC VASO, VAPER, MT-weighted, ABC and ASL fMRI methods all exceed 9 seconds. Despite advancements in acquisition speed, current CBV/CBF-based fMRI techniques remain inadequate for layer-dependent resting-state fMRI. To investigate brain-wide mapping of resting-state functional connectivity, the Nyquist sampling rate per volume should be less than 5 seconds, as resting-state data is typically low-pass filtered at 0.1 Hz.

This study aims to reduce the inter-layer dependency while achieving whole-brain coverage with a volumetric repetition time (TR) under 5 seconds for layer-dependent fMRI. To accomplish this, we revisited the BOLD EPI method due to its rapid acquisition capabilities and explored approaches to reduce inter-layer dependency. Inter-layer dependency in BOLD signals arises from two distinct

sources: extravascular and intravascular components. For the extravascular component, the blooming effect is a major source to inter-layer dependency. This effect primarily stems from large vessels, particularly pial veins, and can extend into remote tissue voxels. Consequently, depth-dependent profiles can be contaminated by these distant pial vessels. To mitigate the blooming effect while maintaining rapid acquisition, we employed a GE-EPI sequence at 3T instead of a 7T scanner. Although transitioning from 7T to 3T penalized BOLD sensitivity, it decreases the susceptibility-induced Larmor frequency shift by 57% and reduces extravascular signal by approximately 35% at 3T (Uludağ et al., 2009 [↗](#)). The reduction in BOLD sensitivity can be partially compensated using NOise Reduction with DIstribution Corrected principal component analysis (NORDIC PCA). (Vizioli et al., 2021 [↗](#)), which has been shown effective in noise removal with minimal spatial blurring (Dowdle et al., 2023 [↗](#)). Furthermore, we employed phase regression method to alleviate the extravascular contamination from large pial veins (Menon, 2002 [↗](#)).

For the intravascular component, addressing the draining-vein issue can involve estimating the depth-dependent BOLD signal through inverse calculations of the leakage model (Havlicek & Uludag, 2020 [↗](#); Heinze et al., 2016 [↗](#)). However, the model-based vein removal methods are primarily exploratory and the effectiveness is still in need of validation (Huber et al., 2021 [↗](#)). To enhance spatial specificity prospectively, small diffusion-weighted gradients, known as velocity-nulling (VN) gradients, can be incorporated into GE-EPI sequences. These gradients suppress signals from moving blood, thereby reducing contributions from pial and draining veins (Boxerman et al., 1995 [↗](#)). Bipolar diffusion gradients have also been used in combination with other sequences, such as SE-BOLD and T2-preparation methods (Duong et al., 2003 [↗](#); Huber, Hua, et al., 2017 [↗](#)). We selected GE-BOLD over SE-BOLD sequences for its better sensitivity, as both SE-BOLD and diffusion-weighted T2-preparation sequences exhibit lower sensitivity. SE-BOLD sensitivity is approximately 20-30% lower than that of GE-BOLD, while diffusion-weighted T2-preparation sequences experience further sensitivity reduction due to incomplete refocusing (Huber, Hua, et al., 2017 [↗](#)).

In this project, we implemented fMRI with 0.9 mm isotropic resolution, VN gradients, and brain-wide coverage at 3T. For volumetric acquisition, we applied the Simultaneous Multislice (SMS) method (Setsompop et al., 2012 [↗](#)). This enabled whole-brain acquisition with 0.9-mm isotropic resolution, a FOV thickness of 113.4 mm along superior-inferior direction, and a TR of 4.0 s. To enhance sensitivity, we employed the NORDIC denoising method, while the phase regression approach was used to suppress macrovascular contributions from pial veins. The strength of the bipolar diffusion gradients, characterized by the b value, was optimized using a button-pressing task. Using the selected b value, we assessed the feasibility of brain-wide layer-dependent functional connectivity (FC) studies at 3T. This was achieved through layer-specific FC mapping with seeds placed in the primary motor cortex (M1) or the primary sensory cortex (S1). Additionally, we conducted a brain-wide layer-dependent connectome analysis using the Shen268 atlas (Shen et al., 2013 [↗](#)). Our findings revealed multiple instances of layer-dependent functional connectivity that were consistent with previously reported results in the literature.

Theory

Draining-vein suppression using velocity-nulling gradient

The BOLD measurement at cortical layers is spatially blurred and biased towards superficial layers due to draining veins, including extravascular and intravascular components (Duong et al., 2003 [↗](#); Turner, 2002 [↗](#)). Here we incorporate a VN gradient into GE-EPI to suppress the fast-flowing intravascular signal from draining veins (see **Figure 1a** [↗](#)). The signal attenuation against the b value will generally follow the exponential decay in the Intravoxel Incoherent Motion (IVIM) process (Le Bihan, 2019 [↗](#)). The IVIM process models the collective motion of water molecules in blood within a vessel network as they transition from one vessel segment to another. This

collective movement can be perceived as a pseudo-diffusion process where average displacements equate to the mean vessel segment length and the mean velocity matches that of the blood in the vessels.

In a capillary network, the average displacement is approximately 100 μm and the mean velocity is around 1 mm/s, yielding a pseudo-diffusion coefficient (D^*) of $10^{-8} \text{ m}^2/\text{s}$ (Le Bihan, 2019 [\[1\]](#)). In the case of cortical penetrating veins, the actual flow velocity is difficult to measure due to their small size. However, optical imaging suggests a mean flow velocity of $\sim 2.5 \text{ mm/s}$ in these veins (Shih et al., 2013 [\[2\]](#)). Assuming that inter-layer dependency arises primarily from the penetrating veins, we approximate an average displacement of 1 mm, based on cortical anatomical studies (Reina-De La Torre et al., 1998 [\[3\]](#)). Accordingly, the pseudo-diffusion coefficient for penetrating veins is $\sim 2.5 \times 10^{-7} \text{ m}^2/\text{s}$. In contrast, the flow velocity in cortical arteries is $\sim 12 \text{ mm/s}$ (Schaffer et al., 2006 [\[4\]](#)), with an average displacement also of 1 mm. This configuration leads to a pseudo-diffusion coefficient of $\sim 1.2 \times 10^{-6} \text{ m}^2/\text{s}$.

The signal amplitude that underlies the effect of the VN gradient follows an exponential decay as represented by the equation $S/S_0 = \exp[-b(D^* + D_{\text{blood}})]$. In this formula, S denotes the vascular signal affected by the VN gradient, S_0 denotes the signal strength without the VN gradient, b denotes the b value of diffusion gradient, D^* indicates the pseudo diffusion coefficient, and D_{blood} indicates the water diffusion coefficient in blood. As the $D^* \gg D_{\text{blood}}$, the signal decay equation can be simplified to $S/S_0 = \exp(-b \cdot D^*)$. Using the predetermined pseudo diffusion coefficients for arteries, veins, and capillaries, we simulated the signal attenuations as illustrated in **Figure 1b** [\[5\]](#). Our results imply that a relatively small b value should effectively suppress the contribution from draining veins, yet simultaneously cause only minimal attenuation of signals originating from capillaries.

Results

Optimization of VN gradient strength

We evaluated the effects of the VN gradient at different b values by examining the depth-dependent BOLD activation profile in the primary motor cortex (M1) during a button-pressing task as shown in **Figure 2a** [\[6\]](#). **Figure 2b** [\[7\]](#) displays the GE-EPI image in three orthogonal views, highlighting the M1 region within a red box, with a zoom-in view and a color-coded layer representation on the right.

The NORDIC denoising method was applied to enhance sensitivity. Recent studies have demonstrated that the NORDIC method effectively removes noise while introducing minimal spatial blurring (Dowdle et al., 2023 [\[8\]](#)). To confirm this in our data, **Figure 3a** [\[9\]](#) and **Figure 3b** [\[10\]](#) compare the BOLD activation maps before and after NORDIC denoising in the visual and motor cortices, respectively. These maps correspond to a TE of 30 ms without phase regression, ensuring the effects of VN gradients and phase regression were controlled. The activation patterns in the denoised maps were consistent with those in the non-denoised maps but showed higher statistical significance. Notably, BOLD activation within M1 was only observed after NORDIC denoising, highlighting the necessity of applying this approach. **Figure 3c** [\[11\]](#) presents the depth-dependent activation profiles in M1, highlighted by the green contours in **Figure 3b** [\[12\]](#). Both profiles exhibited similar trends. However, the non-denoised profile showed larger confidence intervals compared to the NORDIC-denoised profile, as expected. These results confirm that NORDIC denoising enhances sensitivity without distorting the functional signal.

Depth-dependent BOLD activation profiles in M1 were extracted using a voxel-based approach. **Figure 4a-d** [\[13\]](#) illustrate these profiles across different subjects, with each column representing different TEs and b values. Three out of four subjects exhibited double-peak response patterns.

Figure 1

Velocity-nulling (VN) gradient in GE-EPI.

(a) The diagram of pulse sequence. The VN gradient is highlighted in light blue. (b) The simulated signal attenuation against b values.

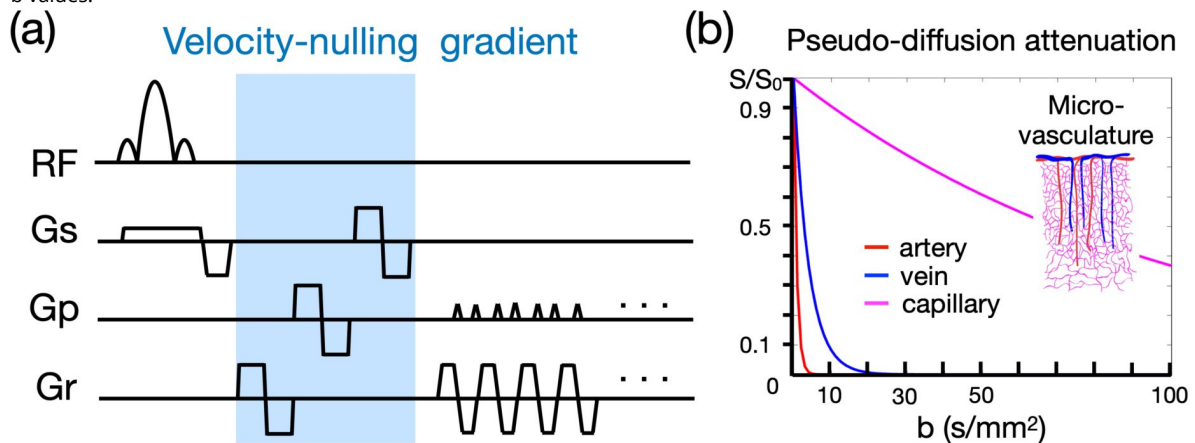
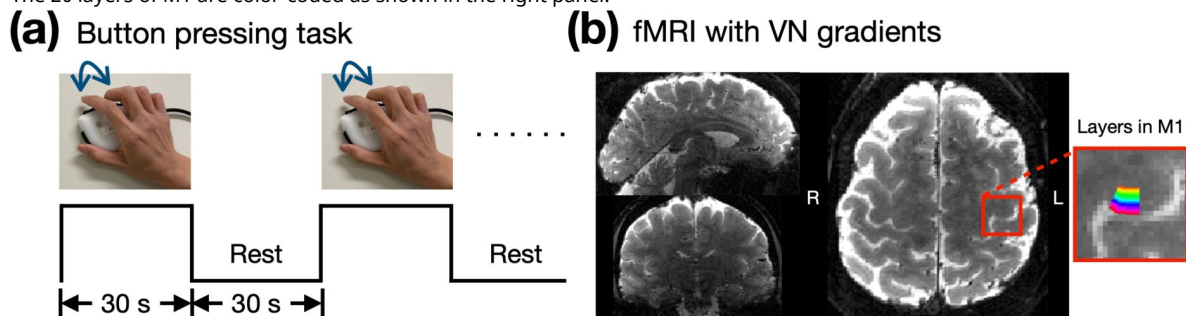


Figure 2

Empirical results of the button-press task showing varying levels of draining-vein suppression in different participants.

(a) The paradigm of button-pressing task. (b) The acquired image with 0.9-mm isotropic resolution in three orthogonal views. The 20 layers of M1 are color-coded as shown in the right panel.



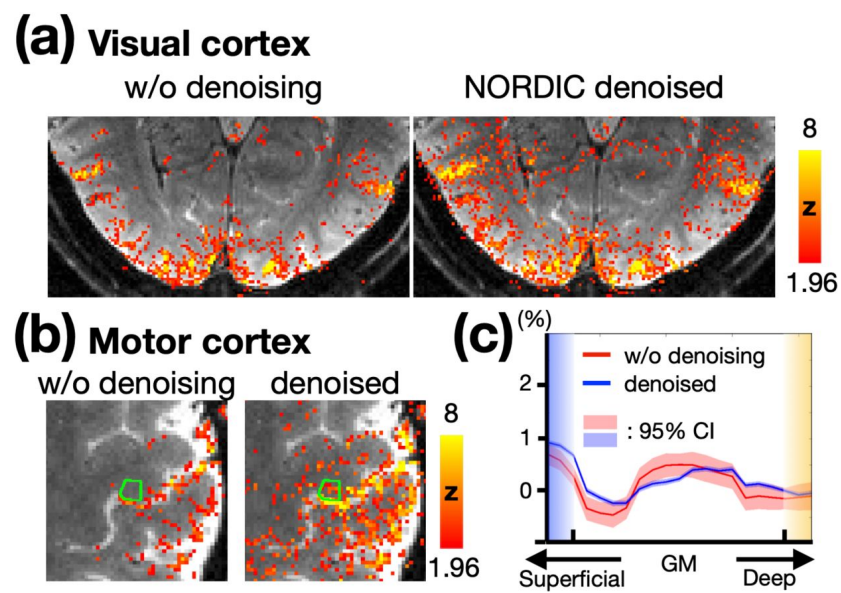


Figure 3

(a) BOLD activation maps in the visual cortex without and with NORDIC denoising. (b) BOLD activation maps in the motor cortex without and with NORDIC denoising. The statistical maps were corrected (uncorrected $p < 0.05$; corrected $p < 0.05$) and color-coded as indicated by the color bar. (c) Depth-dependent BOLD activation profiles for the M1 regions shown in (b). The shaded areas in light red and blue represent the 95% confidence intervals.

Notably, using b value of 8 s/mm^2 appeared to excessively suppress deep-layer activation. Comparing column 2nd and 3rd ($b = 0$ and 6 s/mm^2 , respectively, at $TE = 38 \text{ ms}$) showed that the percentage of BOLD signal change in superficial layers is generally lower with b of 6 s/mm^2 than with b of 0 , suggesting that VN gradient-induced signal suppression is more pronounced in superficial layers. Moreover, **Figure 4a-d** also present layer-dependent profiles with and without phase-regression (PR). Observable differences were noted in only two subjects, as indicated by the yellow arrows in the first column of **Figure 4c** and the second column of **Figure 4d**. Overall, PR primarily influenced the superficial layers within M1, though its impact was not substantial.

To investigate whether the low efficacy of phase regression observed in a small M1 region could be generalized to other brain regions, we evaluated its efficacy in the visual cortex using the same datasets of subject #1. All four parameter combinations were tested. The columns in **Figure 5** represent activation maps under different parameter combinations, while the rows show the maps without phase regression, the maps with phase regression, and the difference maps, from top to bottom. The difference maps demonstrate that phase regression is effective across all tested conditions, at least when $TE \leq 39 \text{ ms}$ and $b \leq 8 \text{ s/mm}^2$. Consistent with the findings in **Figure 4**, the reduction of macrovascular-induced effects was most prominent near the pial surface. As highlighted by the blue circles in **Figure 5**, VN gradients effectively suppressed macrovascular signals as well. Furthermore, unlike the results in **Figure 4**, the reduction in activation strength in the visual cortex was less obvious when using a b value of 8 s/mm^2 . This suggests that the degree of signal reduction associated with b values may vary across brain regions. Based on the findings from **Figure 4** and **Figure 5**, we selected a b value of 7 s/mm^2 as a reasonable compromise and employed it in subsequent experiments.

Assessment of Inter-layer dependency

To investigate the effects of VN gradients on signal dependency across cortical depth, we resampled the fMRI time series, both with and without VN gradients, onto the brain surface at different cortical depths. For each cortical vertex, intra-cortical FC matrices were computed and subsequently averaged across the entire brain surface. As shown in **Figure 6a**, the reduction in layer dependency is most pronounced in the superficial layers. Furthermore, reductions in layer dependency were observed between the superficial and middle layers, as well as between the middle and deep layers, with these regions highlighted by the red circle. These reductions induced by VN gradients were statistically significant (FDR-corrected $p < 0.05$), as illustrated in **Figure 6b**. The observed reductions in layer-dependent FC argue against the possibility that these changes are merely due to layer-independent signal attenuation. If VN gradients attenuated the signal without suppressing cross-layer blurring, the result would be a more uniform decrease in FC across all layers. Instead, the findings suggest that VN gradients selectively influence intra-cortical FC, supporting their role in improving layer-dependent specificity.

Functional connectivity maps using layer-specific seeds in M1

The sensorimotor network has been studied extensively, not only region-specific but also layer-specific characteristics on functional connectivity (Caulier, 1995; Felleman & Van Essen, 1991). In the primary motor cortex, superficial layers are predominantly associated with sensory areas, while deep layers are primarily linked to premotor areas. To evaluate these layer-specific properties in our resting-state data, we conducted the layer-specific FC analysis as illustrated in **Figure 7a**. Superficial and deep layer labels for M1 were generated in EPI space, with volume-based labels derived from individual surface-based M1 labels. The seed time courses were extracted from the superficial and deep layers in M1, and these time courses were used as regressors in the GLM analysis across the brain. The resulting FC maps were resampled onto individual cortical surfaces at various depths and subsequently morphed to a template cortical surface for group-level analysis (refer to the Materials and Methods section for further details).

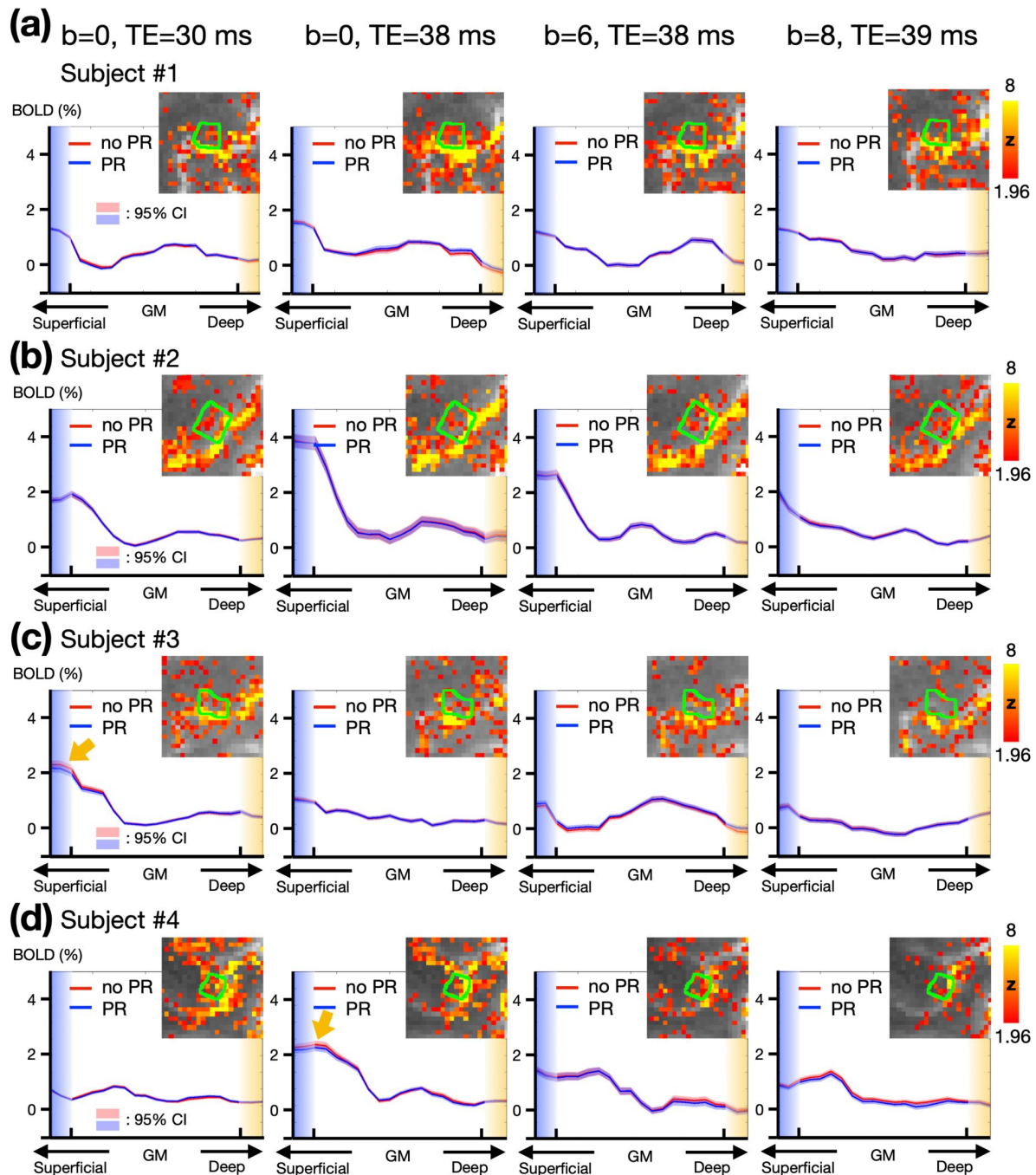


Figure 4

Empirical results of the button-press task showing varying levels of draining-vein suppression in different participants.

(a-d) Depth-dependent BOLD activation profiles in M1. The green contours represent the edges of M1 regions. Each row represents data from a different individual. Columns from left to right correspond to TE = 30 ms, TE = 38 ms, TE = 38 ms with $b = 6 \text{ s/mm}^2$, and TE = 39 ms with $b = 8 \text{ s/mm}^2$. The statistical maps were corrected (uncorrected $p < 0.05$; corrected $p < 0.05$) and color-coded as indicated by the color bar. The shaded areas in light red and blue represent the 95% confidence intervals (CI).

Figure 5

The activation maps correspond to visual cues from the button-pressing task.

The difference maps were generated by subtracting the activation maps in the middle row (with phase regression) from those in the top row (without phase regression). The blue circles highlight the effective suppression of macrovascular signals by VN gradients.

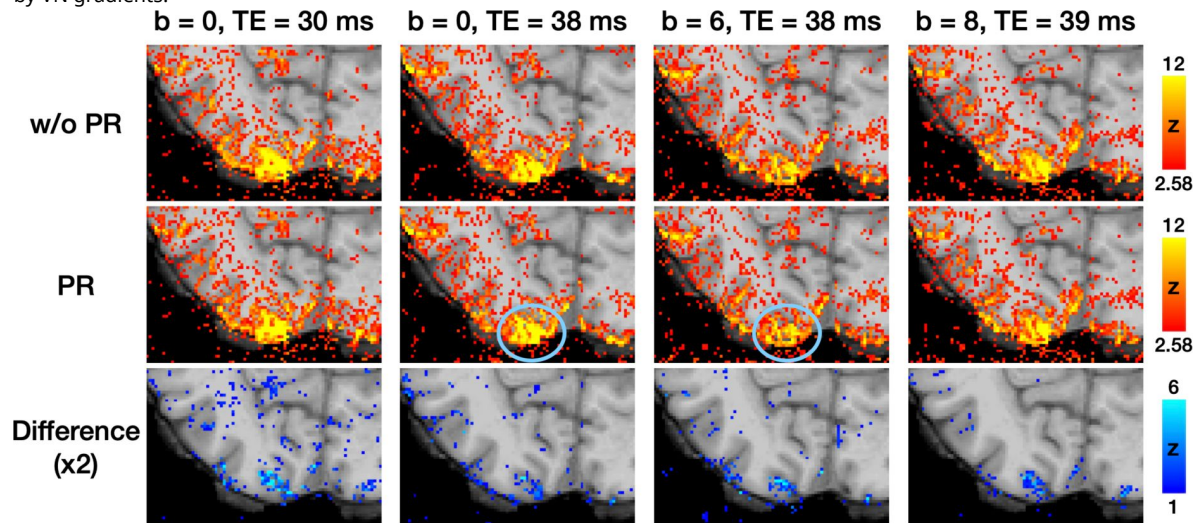
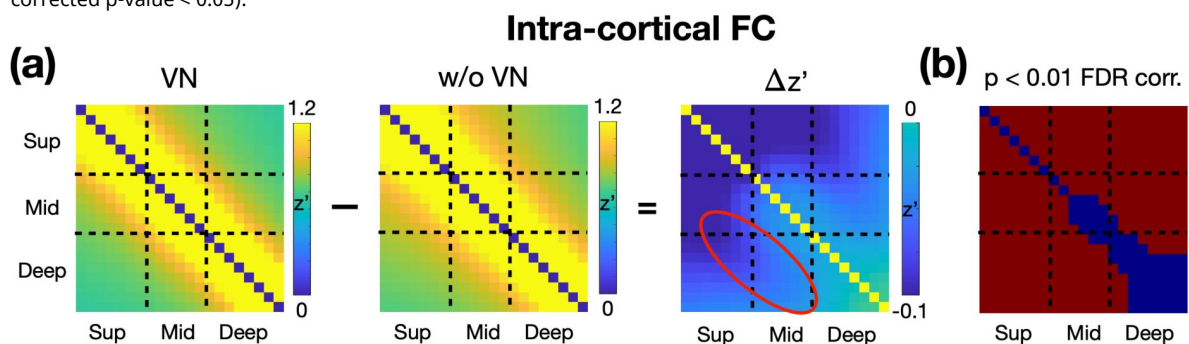


Figure 6

Impact of VN Gradients on Intra-Cortical FC Matrices.

(a) The panels from left to right display the intra-cortical FC matrix derived from VN fMRI, the intra-cortical FC matrix derived from regular fMRI, and the difference between the two matrices. All FC matrices are Fisher's z-transformed. (b) Statistical significance of the differences in the FC matrices. Red regions indicate the differences are statistically significant (FDR-corrected p-value < 0.05).



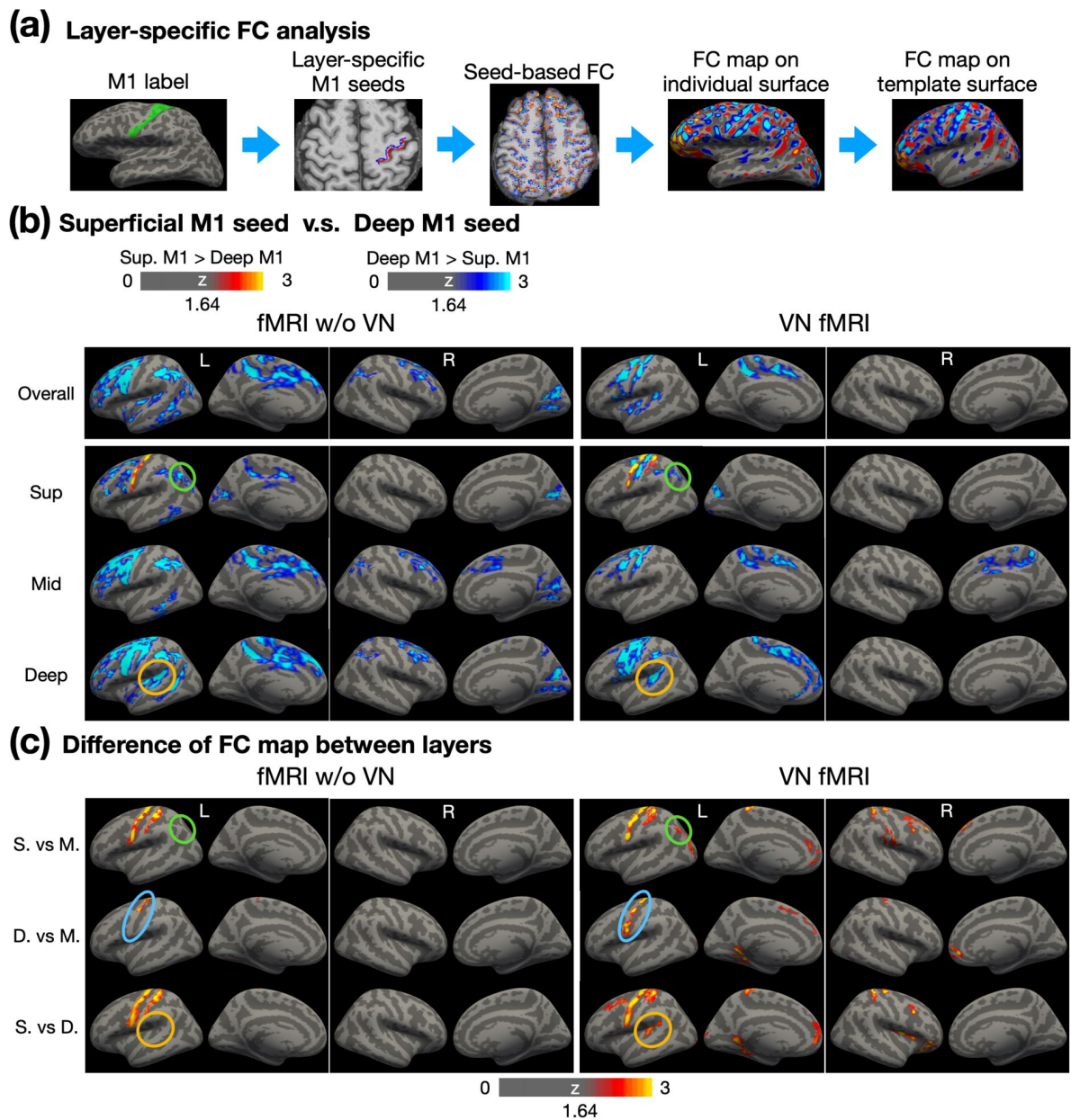


Figure 7

Layer-dependent FC analysis using layer-specific M1 seeds.

(a) Workflow for conducting layer-specific FC analysis. Surface-based labels were converted from individual surface to volume space. Red and blue colors represent the superficial and deep layers, respectively. Seed-based FC analysis was performed in EPI space, followed by morphing to individual surfaces and then to the template surface. (b) Contrast of FC maps corresponding to superficial, middle and deep layers. Warm colors represent stronger FC associated with superficial layers in M1, while cool colors indicate stronger FC associated with deep layers in M1. (c) Differential FC maps showing the magnitude of differences between layers, including superficial vs. middle, deep vs. middle, and superficial vs. deep layers. Abbreviations: Sup – superficial; Mid – middle; S. – superficial; M. – middle; D. – deep.

Figure 7b shows a comparison of FC maps derived from superficial-layer and deep-layer seeds. To evaluate the layer specificity of these FC patterns, **Figure 7c** displays the magnitude of cross-layer differences based on the FC maps shown in **Figure 7b**. In **Figure 7b**, the first row illustrates overall FC across cortical depth, with conventional fMRI results (without VN gradients) shown in the left panel and VN fMRI results in the right panel. The second through fourth rows highlight FC maps for superficial, middle, and deep layers, respectively. Generally, deep M1 exhibited stronger FC compared to superficial M1 within the sensorimotor network and beyond. In addition, the FC patterns obtained from conventional fMRI appeared more spatially diffuse compared to those from VN fMRI, reflecting the latter's improved specificity. The VN fMRI results further revealed that superficial M1 exhibited stronger connectivity with superficial S1, particularly Brodmann area (BA) 1, as shown in the superficial-layer FC maps. This observation aligns with previous reports of connectivity between superficial M1 and sensory regions (Caulier, 1995). The findings in **Figure 7c** further indicate that superficial M1 primarily connects with superficial layers of S1, with relatively weaker connectivity to middle or deep layers.

Figure 7c examines the inter-layer dependency. If cross-layer signals are blurred, such as by contamination from draining veins or the blooming effect, the statistical distinctions across layers in FC maps would diminish. In **Figure 7c**, conventional fMRI demonstrated cross-layer differences only in the M1 and S1 regions. In contrast, VN-enhanced fMRI showed a significantly greater number of regions exhibiting cross-layer differences. For example, as highlighted by the green circles in **Figure 7b**, both conventional and VN-enhanced fMRI showed stronger FC between the angular gyrus (AG) and deep M1 compared to superficial M1. However, while conventional fMRI failed to detect significant differences between superficial and middle layers, VN fMRI identified these differences as statistically significant, as highlighted by green circles in **Figure 7c**.

Furthermore, the deep-layer seed in M1 is expected to exhibit stronger FC with deep layers than with middle layers within the same region. For VN fMRI, this pattern of stronger deep-layer connectivity was observed across nearly the entire M1 in **Figure 7c**, whereas it was restricted to less than half of the M1 in conventional fMRI, as shown by the blue circles. Additionally, the brown circles in **Figure 7b** indicate that deep layers in Wernicke's area exhibited FC with deep M1 in both conventional and VN fMRI. However, as shown by the brown circles in **Figure 7c**, statistical analyses revealed that this connectivity was preferentially associated with the deep layers rather than the superficial layers in VN fMRI. Such a layer-specific distinction was not evident in conventional fMRI, highlighting the enhanced layer specificity of VN fMRI.

To evaluate whether the spatially diffuse FC associated with deep-layer M1 arises from artifacts specifically related to deep-layer signals, we conducted similar FC analyses using a seed region in S1. From a neurophysiological perspective, the S1 region primarily supports bottom-up processing and is expected to exhibit FC that is more confined to the sensorimotor network. If the patterns observed in **Figure 7** were driven by deep-layer-related artifacts, the results for S1 would be expected to demonstrate diffuse FC patterns similarly, given the close anatomical proximity of M1 and S1.

Figure 8a illustrates the S1 seed placement. The left panel shows the surface-based S1 label on the template surface, while the right panel displays superficial and deep-layer S1 labels on an individual brain. Following a similar layout to **Figure 7b**, the top to bottom rows in **Figure 8b** represent the FC maps for overall, superficial, middle, and deep layers, respectively. The FC maps of conventional fMRI appeared to be more spatially diffuse, extending beyond the sensorimotor network. In contrast, the FC maps of VN fMRI are relatively confined to sensorimotor network, consistent with neurophysiological expectations. These findings suggest that the stronger and more widespread FC associated with deep-layer M1 were unlikely due to deep-layer-related artifacts.

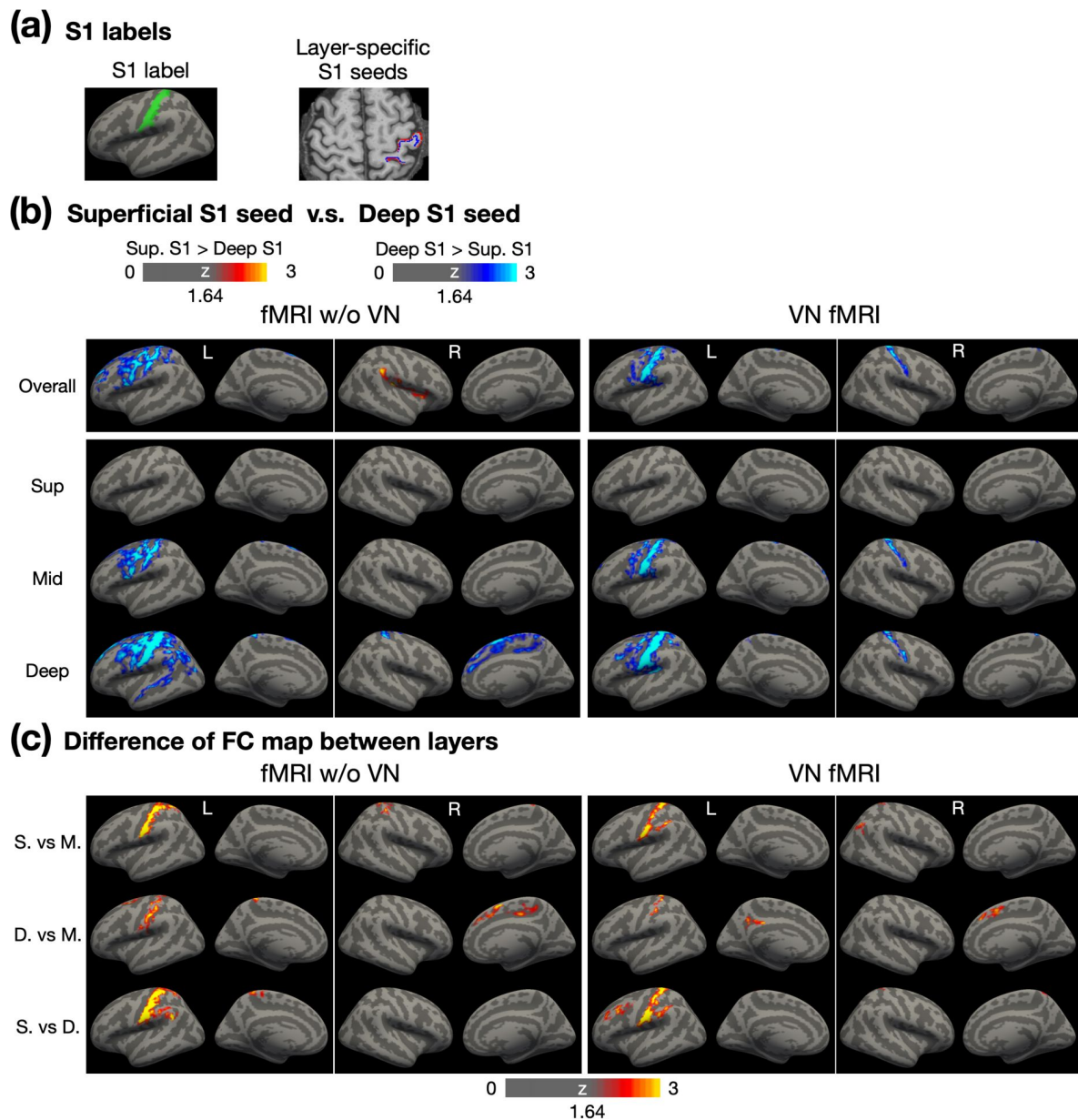


Figure 8

Layer-dependent FC analysis using layer-specific S1 seeds.

(a) The illustration of surface-based and volume-based S1 labels. (b) Contrast of FC maps corresponding to superficial, middle and deep layers. Warm colors represent stronger FC associated with superficial layers in S1, while cool colors indicate stronger FC associated with deep layers in S1. (c) Differential FC maps showing the magnitude of differences between layers, including superficial vs. middle, deep vs. middle, and superficial vs. deep layers. Abbreviations: Sup – superficial; Mid – middle; S. – superficial; M. – middle; D. – deep.

In the FC maps of VN fMRI corresponding to the superficial layer (**Figure 8b**), no significant differences were observed between superficial- and deep-layer FC maps, indicating that the superficial and deep layers in S1 exhibited comparable FC strength across the cerebral cortex. However, in the middle- and deep-layer FC maps, the deep-layer S1 demonstrated stronger FC than the superficial-layer S1, particularly in bilateral S1 and the left S1 and M1 regions. Cross-layer statistical maps in **Figure 8c** further revealed that the deep-layer seed in S1 predominantly connected with the middle and deep layers within S1, rather than with the superficial layers. Notably, other regions showing significantly stronger FC associated with the deep-layer S1 did not exhibit clear layer specificity, potentially due to an insufficient sample size.

Brain-wide layer-specific functional connectivity

To explore brain-wide layer-specific functional connectivity, we parcellated the brain using Shen268 functional atlas (refer to the Materials and Methods section for further details). The group-level statistics for the layer-specific functional connectivity matrices are presented in **Figure 9a**, with ROIs organized according to their respective functional networks. The FC matrix on the left was derived from conventional fMRI data without the application of VN gradients, while the matrix on the right was generated using VN fMRI data. In general, the FC matrix from conventional fMRI displayed more widespread connectivity than that of VN fMRI, consistent with the seed-based FC results shown in **Figure 7** and **Figure 8**.

The enhanced layer specificity provided by VN fMRI enables detailed mapping of the layer-specific brain connectome. Examples from the visual, sensorimotor, and default-mode functional networks are presented in **Figure 9b**, **7c**, and **7d**, respectively. The surface-based labels of ROI pairs are displayed on the left and upper sides of the depth-dependent FC matrices. Focusing on VN fMRI results, within the visual network, the primary visual cortex (V1) exhibited feedback connectivity from the deep layers of the visual area V2 to the superficial layers of V1. In the sensorimotor network, the pre-motor area received inputs exclusively in its superficial layers, originating from both the superficial and deep layers of the S1. Within the default mode network (DMN), the ventral posterior cingulate cortex (vPCC) demonstrated connectivity dominated by its middle layers. Notably, the angular gyrus (AG) exhibited feedforward connectivity to the middle layers of vPCC, consistent with its role as a major hub in the DMN and in alignment with previously reported findings from a laminar-specific fMRI study (Huber et al., 2021a). In contrast, the results from conventional fMRI showed diffuse and less distinct patterns in the depth-dependent FC matrices, diverging from established findings in the literature. This further underscores the capability of VN fMRI in resolving layer-specific connectivity with greater accuracy.

Discussion

To develop brain-wide layer-specific functional connectivity using depth-dependent functional information, several technical criteria must be met: sub-mm spatial resolution, high spatial specificity, adequate temporal resolution, sufficient functional sensitivity, and global brain coverage. In this study, we addressed those challenges by incorporating a number of techniques, such as velocity-nulling gradients, SMS acceleration, phase regression and NORDIC denoising techniques. As a result, we were able to develop a VN fMRI sequence featuring 0.9-mm isotropic spatial resolution, a TR of 4 seconds and brain-wide coverage. The effects of TE and b values associated with the VN gradient were evaluated. Based on fMRI results from a button-pressing task, a b value of 7 s/mm² was selected to achieve a reasonable balance between sensitivity and specificity. Using this parameter, we assessed the validity of layer-specific FC through seed-based analyses. Results revealed distinct FC patterns associated with superficial and deep layers in M1, and the resulting FC maps exhibited significant differences between layers, demonstrating that VN fMRI enhances inter-layer independence. Additional FC analyses were also conducted with a seed

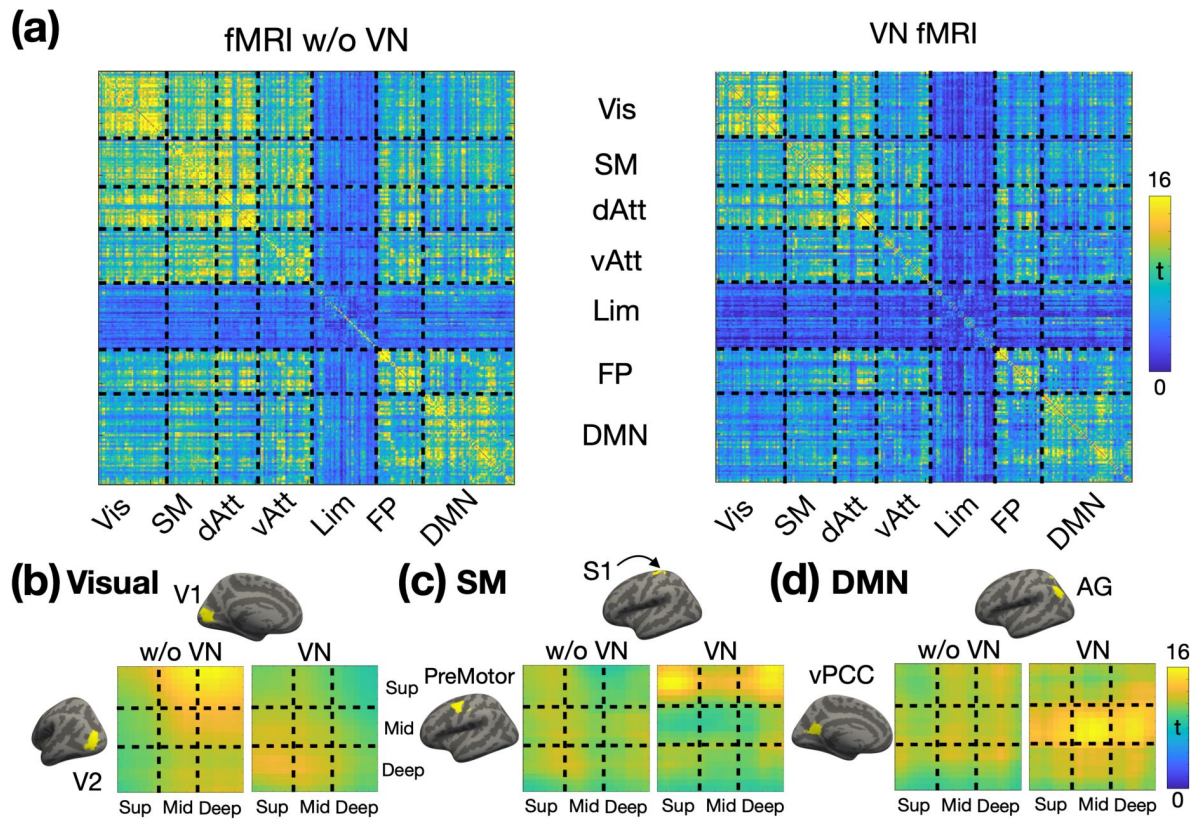


Figure 9

Layer-specific functional connectivity across functional networks (N = 14).

(a) Fisher's z transformed FC matrices. The FC matrix on the left represents the Fisher's z values without a VN gradient, while the matrix on the right corresponds to the Fisher's z values with a VN gradient. From (b) to (d) displays the depth-dependent FC matrices for a representative ROI pair in visual, sensorimotor and default-mode networks respectively. T-values are color-coded as indicated by the color bar. The left panel displays the matrix without a VN gradient, and the right panel presents the matrix with a VN gradient. Abbreviations: V1 – primary visual cortex; V2 – secondary visual cortex; S1 – primary sensory cortex; vPCC – ventral posterior cingulate cortex; AG – angular gyrus; Vis – visual network; SM – sensorimotor network; dAtt – dorsal attention network; vAtt – ventral attention network; Lim – limbic network; FP – frontoparietal network; DMN – default-mode network.

placed in S1 to further validate the findings. Moreover, brain-wide layer-dependent FC analyses revealed several findings consistent with existing literature, supporting the efficacy and reliability of the VN fMRI approach in resolving layer-dependent functional connectivity.

The proposed VN-fMRI method employs VN gradients to selectively suppress signals from fast-flowing blood in large vessels. Although this approach may initially appear to diverge from the principles of CBV-based techniques (Chai et al., 2020 [↗](#); Huber, Handwerker, et al., 2017 [↗](#); Pfaffenrot & Koopmans, 2022 [↗](#); Priovoulos et al., 2023 [↗](#)), which enhance sensitivity to vascular changes in arterioles, capillaries, and venules while attenuating signals from static tissue and large veins, it aligns with the fundamental objective of all layer-specific fMRI methods. Specifically, these approaches aim to maximize spatial specificity by preserving signals proximal to neural activation sites and minimizing contributions from distal sources, irrespective of whether the signals are intra- or extra-vascular in origin. In the context of intravascular signals, CBV-based methods preferentially enhance sensitivity to functional changes in small vessels (proximal components) while demonstrating reduced sensitivity to functional changes in large vessels (distal components). For extravascular signals, functional changes are a mixture of proximal and distal influences. While tissue oxygenation near neural activation sites represents a proximal contribution, extravascular signal contamination from large pial veins reflects distal effects that are spatially remote from the site of neuronal activity. CBV-based techniques mitigate this challenge by unselectively suppressing signals from static tissues, thereby highlighting contributions from small vessels. In contrast, the VN fMRI method employs a targeted suppression strategy, selectively attenuating signals from large vessels (distal components) while preserving those from small vessels (proximal components). Furthermore, the use of a 3T scanner and the inclusion of phase regression in the VN approach mitigates contamination from large pial veins (distal components) while preserving signals reflecting local tissue oxygenation (proximal components). By integrating these mechanisms, VN fMRI improves spatial specificity, minimizing both intravascular and extravascular contributions that are distal to neuronal activation sites.

Bipolar diffusion gradients have been employed to nullify signals from fast-flowing blood, as demonstrated by (Boxerman et al., 1995 [↗](#)). Their work showed that vessels with flow velocities producing phase changes greater than π radians due to the bipolar gradients experience significant signal attenuation. The critical velocity for such attenuation can be calculated using the formula: $1/(2\psi G\Delta)$ where ψ is the gyromagnetic ratio, G is the gradient strength, δ is the gradient pulse width and Δ is the time between the two bipolar gradient pulses. In the framework of Boxerman et al. at 1.5T, the critical velocity for b value of 10 s/mm^2 is $\sim 8 \text{ mm/s}$, resulting in a $\sim 30\%$ reduction in functional signal. In our 3T study, b values of 6, 7, and 8 s/mm^2 correspond to critical velocities of 16.8, 15.2, and 13.9 mm/s , respectively. The flow velocities in capillaries and most venules remain well below these thresholds. Notably, in our VN fMRI sequences, bipolar gradients were applied in all three orthogonal directions, whereas in Boxerman et al.'s study, the gradients were applied only in the z -direction. Given the voxel dimensions of $3 \times 3 \times 7 \text{ mm}^3$ in the 1.5T study, vessels within a large voxel are likely oriented in multiple directions, meaning that only a subset of fast-flowing signals would be attenuated. Therefore, our approach is expected to induce greater signal reduction, even at the same b values as those used in Boxerman et al.'s study.

Bipolar diffusion gradients have also been employed in layer-dependent fMRI at 7T, but in conjunction with T2 preparation methods (Huber, Hua, et al., 2017 [↗](#)) rather rapid GE-EPI sequences. This is probably due to the limited TE associated with GE-EPI at 7T, which is insufficient to accommodate bipolar diffusion gradients in an ultra-high-resolution regime. However, it is noteworthy that the functional sensitivity of diffusion-weighted T2 preparation methods is considerably lower than the GE-BOLD method (Huber et al., 2021b [↗](#)). While higher magnetic field strengths could enhance sensitivity, the extravascular effect in GE-BOLD method decreases the spatial specificity. The implementation of VN fMRI at 3T not only addresses the extravascular effect but also leverages several unique advantages of 3T over 7T. First, the extended $T2^*$ values at 3T lead to diminished $T2^*$ blurring along the phase-encoding direction, thus yielding a more

focused point-spread function (PSF). Second, geometric distortions and signal losses due to susceptibility effects are less prominent at 3T relative to those at 7T. Third, the feasibility of conducting layer-dependent fMRI at 3T broadens the scope for multi-site, large-scale investigations. These attributes are of critical importance for a broad range of neuroscience applications.

In addition to draining-vein contamination, another source of cross-layer blurring is extravascular effect which commonly originates from large pial veins. According to the biophysical models (Ogawa et al., 1993 [\[1\]](#)), the extravascular contamination from the pial surface is inversely proportional to the square of the distance from vessel. For a vessel diameter of 0.3 mm and an isotropic voxel size of 0.9 mm, the induced frequency shift is reduced by at least 36-fold at the next voxel. Notably, a vessel diameter of 0.3 mm is larger than most pial vessels. Theoretically, the extravascular effect contributes minimally to inter-layer dependency, particularly at 3T compared to 7T due to weaker susceptibility-related effects at lower field strengths. Empirically, as shown in **Figure 7c** [\[1\]](#), the results at M1 demonstrated that layer specificity can be achieved statistically with the application of VN gradients.

Although extravascular effects have minimal impact on signals from the middle and deep cortical layers (Markuerkiaga et al., 2016 [\[2\]](#)), signal contamination from pial veins can still affect the superficial layers. This study addressed this issue by employing phase regression, a method used to reduce biases toward signals from superficial cortical layers (Knudsen et al., 2023 [\[3\]](#); Koopmans et al., 2010 [\[4\]](#); Markuerkiaga et al., 2021 [\[5\]](#); Scheeringa et al., 2016 [\[6\]](#)). Similarly, our findings in **Figure 4** [\[1\]](#) and **Figure 5** [\[1\]](#) demonstrate that phase regression effectively suppresses macrovascular contributions primarily near the gray matter/CSF boundary, regardless of whether VN gradients are applied. These results suggest that phase regression can complement VN gradients, offering a robust strategy to enhance spatial specificity in layer-specific fMRI.

The distinction between FC maps associated with superficial- and deep-layer seeds was used to evaluate the reliability of layer-dependent resting-state FC. In the work of (Huber, Handwerker, et al., 2017 [\[7\]](#)), the superficial- and deep-layer seeds at M1 were defined manually and the corresponding FC maps were restricted to the same 2D image plane as the seeds. Their results showed that superficial M1 exhibited FC with a region in the post-central gyrus, while deep-layer M1 connected with a region in the pre-central gyrus. In our study, superficial- and deep-layer seeds in M1 were generated automatically using predefined anatomical labels from FreeSurfer. The resulting FC maps were consistent with previous findings, showing that superficial-layer M1 functionally connected with post-central regions, while deep-layer M1 exhibited FC with pre-central regions. Additionally, our results revealed that deep-layer M1 was functionally connected to the deep layers of Wernicke's area. As part of the language network (Tomasi & Volkow, 2012 [\[8\]](#)) and primarily involved in language comprehension (Johns, 2014 [\[9\]](#)), Wernicke's area likely integrates information through long-range corticocortical projections (Huber et al., 2021b [\[10\]](#); Lawrence et al., 2019 [\[11\]](#)). Moreover, we observed that deep-layer M1 connected with the superficial layer of the dorsal angular gyrus (AG), a critical hub in the default-mode network (Biswal et al., 1995 [\[12\]](#); Biswal, 2012 [\[13\]](#)). This connectivity suggests that M1 and AG are not functionally isolated even during rest. The deep-to-superficial connectivity implies that motor systems may provide contextual signals to high-order cognitive regions, potentially via a feedback pathway. These findings suggested the integrative role of M1 in facilitating communication across functional networks during resting state.

The analysis of brain-wide layer-dependent FC is a cutting-edge approach that requires sufficient spatial specificity, brain coverage, sensitivity, and sampling rates to be feasible. Some of the previously mentioned methods have investigated layer-dependent FC within specific networks, such as the visual network (Koiso et al., 2022 [\[14\]](#)) or the cortico-thalamic network (Deshpande et al., 2022 [\[15\]](#)). The MAGEC VASO method (Huber et al., 2021b [\[10\]](#)) further extended this exploration to brain-wide mapping of the functional human connectome using the Shen268 atlas (Shen et al.,

2013) which was also employed in our study. Several reported inter-regional layer-dependent FC patterns are consistent with our findings. However, the FOV in those studies restricts coverage of temporal lobe, which is critical to memory processing and associated with disorders such as Alzheimer's disease. In contrast, our method provides broader brain coverage, making it more suitable for comprehensive layer-dependent FC mapping of the human connectome. This enhanced coverage allows the inclusion of regions critical for understanding large-scale brain networks and their roles in both health and disease.

Recent advancements in ultrahigh-resolution fMRI have enabled the exploration of brain-wide layer-specific FC across the human brain (see **Table 1** for details). These imaging methods are broadly categorized as either BOLD-based or CBV-based. Here we defined brain-wide acquisition as imaging protocols that cover more than half of the human brain, specifically >55 mm along the superior-inferior axis. Among the BOLD-based approaches, the EPI with keyhole (EPIK) method (Yun et al., 2022) enhances spatial resolution by fully sampling the central portion of k-space for every volume while sparsely sampling the outer portions. The EPIK method has demonstrated functional sensitivity by identifying several resting-state functional networks. Another BOLD-based technique has been employed to investigate layer-dependent cortico-thalamic FC and interhemispheric cortico-cortical FC (Deshpande et al., 2022). However, due to draining-vein contamination and lower resolution along the z-axis, the point spread function remains relatively flat, and the results require further experimental validation (Deshpande et al., 2022). For CBV-based methods, the spatial resolution achieves submillimeter precision along all three orthogonal axes (Chai et al., 2024; Huber et al., 2023; Koiso et al., 2022) with a maximum spatial coverage of 100.8 mm along the z-axis. However, the temporal resolution for CBV-based methods exceeds 5 seconds, raising concerns about temporal aliasing of resting-state signals. Although our method may not be as robust as CBV-based methods at the individual level, it successfully demonstrates layer specificity at the group level with a sample size of $N = 14$.

Limitations

Despite the unprecedented features of our VN fMRI, this study has some limitations. First, VN fMRI method is inherently sensitive to T2*-related contamination. According to the results in **Figure 4**, stable activation in M1 was observed at the single-subject level across most scan protocols. Yet, the layer-dependent activation profiles in M1 were spatially unstable, irrespective of the application of VN gradients. This spatial instability is not entirely unexpected, as T2*-based contrast is inherently sensitive to various factors that perturb the magnetic field, such as eye movements, respiration, and macrovascular signal fluctuations. Furthermore, ICA-based artifact removal was intentionally omitted in **Figure 4** to ensure fair comparisons between protocols, leaving residual artifacts unaddressed. Inconsistency in performing the button-pressing task across sessions may also have contributed to the observed variability. These results suggest that submillimeter-resolution fMRI may not yet be suitable for reliable individual-level layer-dependent functional mapping, unless group-level statistics are incorporated to enhance robustness.

Additionally, the VN gradients may not sufficiently suppress the distal contributions of BOLD signals if the flow velocity in draining veins is relatively slow. Since penetrating veins are a potential source of cross-layer blurring, the flow velocity plays a critical role in the effectiveness of VN gradients. The flow velocity of draining veins is positively correlated with vessel diameter. For vessels with diameters exceeding 0.1 mm, flow velocities can exceed 15 mm/s (Linninger et al., 2013), which allows them to be largely suppressed by VN gradients. However, for vessels with diameters between 0.05 and 0.1 mm, flow velocities typically range from 5 to 15 mm/s, depending on vascular morphology. VN gradients may only partially suppress signals from veins within this size range. Although the signal contribution from individual vessels of this size may be negligible, the cumulative effect of multiple such vessels may become significant. Given that vessel density

Method	Contrast	TR	Coverage along z	Resolution	Field strength	Reference
VASO	CBV	5.2s for VASO + 5.1s for BOLD; 10.3s effectively	94.08 mm	0.84 mm isotropic	7T	(Koiso et al., 2022)
MAGE VASO	CBV	14.137 s (4 shots; 3.5 s per shot)	100.8 mm	0.82×0.82×0.84 mm ³	3T	(L. Huber et al., 2023)
EPIK	BOLD	3.5s	108 mm	0.51×0.51×1.0 mm ³	7T	(Yun et al., 2022)
Regular EPI	BOLD	3 s	55.5 mm	0.85×0.85×1.5 mm ³	7T	(Deshpande et al., 2022)
3D VAPER	CBV/CBF	6 s per volume; 12s effectively	80.64 mm	0.8×0.8×0.84 mm ³	7T	(Chai et al., 2024)
VN fMRI	BOLD	4 s	113.4 mm	0.9×0.9×0.9 mm ³	3T	This study

Table 1

List of brain-wide layer-dependent fMRI methods.

The imaging protocols need to cover more than half of the human brain to be classified as brain-wide acquisitions.

increases linearly from deep to superficial cortical layers (Markuerkiaga et al., 2016 [↗](#)), residual signals may persist even after the application of VN gradients. Increasing the strength of VN gradients could mitigate this issue but might result in sensitivity loss.

Moreover, while our VN fMRI acquisition reduced the TR to 4.0 seconds, this duration remains suboptimal for studies utilizing event-related task designs. Future advancements in acceleration techniques would be highly beneficial for exploring high-level cognitive processes with greater temporal precision. However, such developments fall beyond the scope of this study.

Conclusion

In summary, the developed VN fMRI exhibited layer specificity successfully at 3T. Leveraging its brain-wide coverage and reasonable scan time, the VN fMRI yielded promising results in the study of layer-specific FC. Given the widespread accessibility of 3T scanners, the potential impact of this development is expected to be extensive across various domains of neuroscience research.

Materials and Methods

Participants

In this study, two cohorts of healthy adults were recruited. The first cohort consisted of four participants (3 males, age = 34.3 ± 9.2 years) for motor-task experiment. The second cohort was recruited for resting-state functional connectivity analysis and initially included 16 participants. Following the exclusion of two participants due to excessive head movement, the final sample for the second cohort comprised 14 participants (8 males, age = 30.0 ± 6.2 years). All participants underwent screening to confirm no history of neurological or psychiatric conditions, previous head trauma, or MRI contraindications. Only right-handed individuals are included in this study. Prior to participation, each individual provided informed consent in accordance with the experimental protocol approved by the University of North Carolina at Chapel Hill Institutional Review Board.

Stimulation paradigms

All visual stimuli across the paradigms were presented using PsychoPy software (Version 2022.2.4) (Brainard, 1997 [↗](#)). The stimuli were displayed on a screen positioned at the head end of the magnet bore, and participants viewed the visual presentations through a mirror mounted on the head coil.

Button-pressing task

In a button-pressing task, the primary motor cortex (M1) receives incoming sensory and associative information in its superficial layers. This activation subsequently propagates to the deeper layers, where output signals are generated to ultimately control finger muscles. Consequently, both the superficial and deep layers of M1 exhibit activation, while the middle layers typically do not. Such a double-peak response pattern at M1 has been used as a hallmark to evaluate the spatial specificity in layer-dependent fMRI studies (Chai et al., 2020 [↗](#); Huber, Handwerker, et al., 2017 [↗](#); Knudsen et al., 2023 [↗](#); Priovoulos et al., 2023 [↗](#)).

During the experiment, participants were given a button box and instructed to press the button with index and middle fingers of their right hand. The timing and frequency of the button pressing was synchronized with a video displayed on a screen within the scanner. The video consisted of a

block-designed paradigm, as illustrated in **Figure 2a** [↗](#). The button-pressing frequency was 2 Hz with each ‘ON’ block spanning 30 seconds, immediately followed by a 30-second rest during the ‘OFF’ block.

Common protocol in MR acquisitions

MR images were acquired using a Siemens 3T Prisma scanner (Siemens Healthcare, Erlangen, Germany) and a 32-channel head coil at the Biomedical Research Imaging Center (BRIC) at the University of North Carolina at Chapel Hill. For each participant, an MPRAGE image was acquired for structural imaging using the following imaging parameters: 0.8-mm isotropic resolution, TR/TE/TI = 2400/2.24/1060 ms, flip angle = 8°, in-plane acceleration factor = 2, partition thickness = 0.8 mm, 208 partitions, sagittal slicing, image matrix = 320 × 300, and FOV = 25.6 cm × 24.0 cm.

For functional imaging, the customized SMS sequence with VN gradient was implemented in the vendor-provided IDEA environment (VE11E). The functional data were acquired using a blipped-controlled aliasing in parallel imaging (blipped-CAIPI) SMS imaging ([Setsompop et al., 2012](#) [↗](#)) with two-dimensional (2D) single-shot EPI readout. While imaging protocols varied slightly across different experiments, they all shared the following parameters: isotropic spatial resolution of 0.9 mm; axial slicing; in-plane image dimensions of 224×210; frequency and phase encoding in the left-right and anterior-posterior directions, respectively; 126 slices; SMS factor of 3, and an in-plane acceleration rate of 3. Additionally, a brief fMRI acquisition with an opposite phase-encoding direction was acquired immediately before the functional sessions for distortion correction.

Imaging protocols for task-based functional MRI

A button-pressing task was employed to assess BOLD sensitivity and spatial specificity. To identify the optimal b-value that effectively suppresses fast-flowing intravascular signals in large veins while minimizing signal reduction in localized capillaries and venules, empirical tests were performed using four parameter combinations: 1) b = 0, TE = 30 ms; 2) b = 0, TE = 38 ms; 3) b = 6, TE = 38 ms; and 4) b = 8, TE = 39 ms. A total of 270 volumes were acquired, and the button-pressing task included 18 button-pressing blocks.

Imaging protocols for resting-state functional MRI

Each participant in resting-state fMRI cohort underwent two resting-state fMRI sessions. Participants were instructed to stay motionless, keep their eyes closed, and avoid falling asleep. One of the sessions employed the VN gradient with a b value of 7, with the order of the two sessions being randomized among participants. For each session, a total of 300 volumes were acquired, taking approximately 21 minutes.

Data Preprocessing

The SMS-accelerated EPI time series were reconstructed using in-house MATLAB code, which performed slice GeneRalized Autocalibrating Partial Parallel Acquisition (slice-GRAPPA) ([Setsompop et al., 2012](#) [↗](#)) and in-plane GRAPPA ([Griswold et al., 2002](#) [↗](#)) jointly in one step. The multi-channel reconstructed images were subsequently combined using the adaptive combination method ([Walsh et al., 2000](#) [↗](#)). The reconstructed images were denoised using NORDIC method with g-factor map (https://github.com/SteenMoeller/NORDIC_Raw [↗](#); function name: ‘NIFTI_NORDIC’). The g-factor map was estimated from the image time series. The input images were complex-valued. The width of the smoothing filter for the phase was set as 10, while all other hyperparameters retained their default values. After NORDIC denoising, the time-series data were motion corrected using FSL, slice-timing corrected using sinc interpolation, and distortion corrected using FSL TOPUP ([Smith et al., 2004](#) [↗](#)). For artifact removal, we decomposed the magnitude image of the corrected data into a number of independent components by MELODIC ([Beckmann et al., 2005](#) [↗](#)). With the 28 resting-state runs acquired from 14 participants, we trained the ICA-based denoising tool FIX to auto-classify ICA components into signal and noise

components (Salimi-Khorshidi et al., 2014 [↗](#)). For resting-state fMRI datasets, auto-identified noise components were removed. For task-based fMRI datasets, however, artifact removal was not applied to ensure that only one experimental factor (b value or TE) differed between datasets, while all other preprocessing steps remained consistent. Finally, the time-series were band-pass filtered between 0.01 and 0.1 Hz for resting-state fMRI and high-pass filtered at 0.01 Hz for task-based fMRI.

For task-based fMRI, additional coregistration was performed across sessions to enable comparisons between the four parameter combinations associated with the sessions. Filtered images from each session were temporally averaged and concatenated across sessions to create a unified dataset. The concatenated image series underwent motion correction using FSL's 'mcflirt' tool, resulting in the generation of a mean image across sessions. Subsequently, the filtered time series from all sessions were coregistered to this cross-session mean image using a linear transformation. This approach introduced interpolation errors across all the sessions, ensuring a fair and unbiased comparison.

Phase regression

The phase regression approach was applied to remove the macrovascular component in the BOLD signal as indicated in previous reports (Knudsen et al., 2023 [↗](#); Menon, 2002 [↗](#)). This macrovascular component not only contributes to alterations in signal magnitude, but also introduces phase variations. By utilizing a linear regression approach between the time series of signal magnitude and phase, the macrovascular-associated contribution to the BOLD signal could be suppressed. Prior to phase regression, the time series of real and imaginary components were subjected to motion correction, followed by phase unwrapping. The phase regression was incorporated early in the data processing pipeline to minimize the discrepancy in data processing between magnitude and phase images (Stanley et al., 2021 [↗](#)).

Generation of a layered hand-knob label in M1

For depth-dependent analysis of the button-pressing task, the event-related responses were analyzed using FSL FEAT (Woolrich et al., 2001 [↗](#), 2004 [↗](#)). Based on the BOLD activation maps, the hand knob region of the M1 was identified for each individual using the following criteria: 1) the hand knob region was required to be anatomically located in the precentral sulcus or gyrus; 2) it needed to exhibit consistent BOLD activation across the majority of testing conditions; and 3) the region was expected to show BOLD activation in the deep cortical layers under the condition of $b = 0$ and $TE = 30$ ms. Once the boundaries across cortical depth were defined, the gray matter boundaries of hand knob region were delineated based on the T1-weighted anatomical image and the cortical ribbon mask but excluded the BOLD activation map to minimize potential bias in manual delineation.

The layer-dependent BOLD analysis involves the subsampling along cortical depth within the hand knob region. In this study, we generally follow the pipeline suggested by Dr. Huber (<https://layerfmri.com/2018/03/11/quick-layering/#more-531> [↗](#)). The image slice containing M1 was up-sampled by a factor of 5 using nearest-neighbor interpolation, achieving an in-plane resolution of 0.18 mm isotropic. With the created hand-knob label, we employed the LayNii software tool to form 20 equidistant layers (Huber et al., 2021 [↗](#)). The number 20 was determined according to the suggestion by LayNii that the number of layers may be set to at least 4 times larger than the resolution. Depth-dependent functional time series were extracted and averaged within each layer. The averaged layer-specific time series were then analyzed using a general linear model (GLM). The GLM utilized the double gamma function as the hemodynamic response function (HRF) (Friston et al., 1998 [↗](#)) to model event-related responses.

Resting-state functional connectivity with seeds at different layers

To assess inter-layer dependencies in resting-state data, seeds were generated in the superficial and deep layers of the primary motor cortex (M1) and primary sensory cortex (S1). Surface-based labels, obtained from FreeSurfer, were converted to volume-based labels using the `mri_label2vol` function. For the deep-layer seed, the projection fraction (`proj`) was set to 0.1, while for the superficial-layer seed, it was set to 0.9. The resulting volume-based label images were transformed from anatomical to functional image to enable FC analysis in native EPI space. Seed time courses were calculated by averaging the signal across all voxels within each layer-specific label. These time courses were then used as regressors in a general linear model (GLM) implemented in FSL FEAT to generate FC maps.

To extract layer-specific FC values, the superficial, middle, and deep layers were resampled from EPI space onto individual cortical surfaces using `mri_surf2vol` with the following projection options: ‘`projfrac-avg 0.85 0.9 0.05`’, ‘`projfrac-avg 0.45 0.5 0.05`’, ‘`projfrac-avg 0.1 0.15 0.05`’ respectively. For overall FC values across cortical depth, the projection range ‘`projfrac-avg 0.1 0.9 0.05`’ was applied. The resulting surface-based FC maps were morphed from individual brain surfaces to a template brain using `mri_surf2surf`. Statistical significance of the group-averaged Fisher’s z-transformed FC values was assessed using a t-test with the FreeSurfer command ‘`mri_glmfit`’. Multiple comparison correction was performed using a Monte Carlo simulation implemented in FreeSurfer, with 4,000 permutations. The vertex-wise and cluster-wise p-value thresholds were set to 0.05. The corrected p-values were then converted back to z-scores for better visualization.

Surface-based layer-specific functional connectivity analysis

For surface-based subsampling along the cortical depth, we linearly coregistered the individual T1 volume onto the time-averaged EPI volume using Advanced Normalization Tools (ANTs; <http://stnava.github.io/ANTs/>). Following this, we resampled the processed EPI volumes onto the cortical surface at different cortical depths using the FreeSurfer command ‘`mri_vol2surf`’. Adapting the layering strategy in volume space, we converted 20 cortical layers from volume to surface space. Here, the cortical depth ranged from -0.125 to 1.0625, with 0 and 1 representing the GM/WM and CSF/GM boundaries, respectively. Surface-based spatial smoothing was applied with full-width half-magnitude (FWHM) of 3 mm.

In order to investigate layer-specific functional connectivity throughout the brain, we employed the Shen268 functional parcellation (Shen et al., 2013). The volume-based Shen268 parcellation was converted onto the cortical surface as shown in **Figure 10a**. The volumetric EPI time-series for each individual were resampled onto their individual brain surfaces before being projected onto the template brain surface. The template surface is of ‘`fsaverage5`’ from FreeSurfer. We also categorized the parcels in Shen268 atlas into 7 functional networks using the functional atlas reported by Yeo et al. (Yeo et al., 2011). The time course associated with each region-of-interest (ROI) was obtained by averaging all time courses within the ROI. Depth-dependent functional connectivity was calculated using Pearson correlation and then converted into Fisher’s z values, as illustrated in **Figure 10b**. Group-level analyses were performed using one-sample t-tests on the Fisher’s z-transformed correlation matrices.

Acknowledgements

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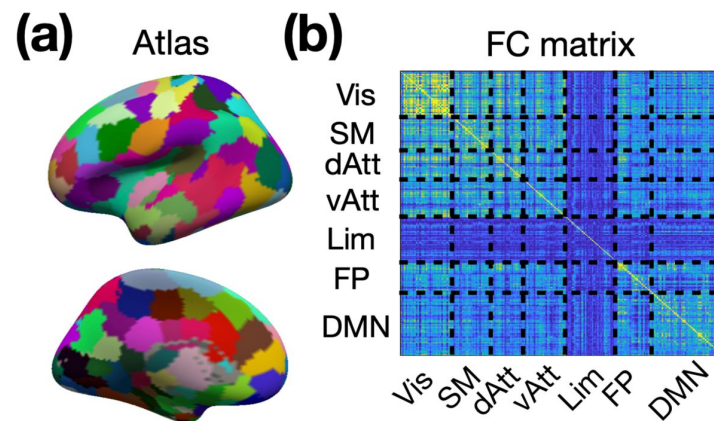


Figure 10

Generation of layer-specific functional connectivity.

(a) The surface-based Shen268 functional parcellation. (b) The depth-dependent functional connectivity matrix. Abbreviations: Vis – visual network; SM – sensorimotor network; dAtt – dorsal attention network; vAtt – ventral attention network; Lim – limbic network; FP – frontoparietal network; DMN – default-mode network.

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Reviewer #1 (Public review):

Summary:

This study aims to provide imaging methods for users of the field of human layer-fMRI. This is an emerging field with 240 papers published so far. Different than implied in the manuscript, 3T is well represented among those papers. E.g. see the papers below that are not cited in the manuscript. Thus, the claim on the impact of developing 3T methodology for wider dissemination is not justified. Specifically, because some of the previous papers perform whole brain layer-fMRI (also at 3T) in more efficient, and more established procedures.

The authors implemented a sequence with lots of nice features. Including their own SMS EPI, diffusion bipolar pulses, eye-saturation bands, and they built their own reconstruction around it. This is not trivial. Only a few labs around the world have this level of engineering expertise. I applaud this technical achievement. However, I doubt that any of this is the right tool for layer-fMRI, nor does it represent an advancement for the field. In the thermal noise dominated regime of sub-millimeter fMRI (especially at 3T) it is established to use 3D readouts over 2D (SMS) readouts. While it is not trivial to implement SMS, the vendor implementations (as well as the CMRR and MGH implementations) are most widely applied across the majority of current fMRI studies already. The author's work on this does not serve any previous shortcomings in the field.

The mechanism to use bi-polar gradients to increase the localization specificity is doubtful to me. In my understanding, killing the intra-vascular BOLD should make it less specific. Also, the empirical data do not suggest a higher localization specificity to me.

Embedding this work in the literature of previous methods is incomplete. Recent trends of vessel signal manipulation with ABC or VAPER are not mentioned. Comparisons with VASO are outdated and incorrect.

The reproducibility of the methods and the result is doubtful (see below).

I don't think that this manuscript is in the top 50% of the 240 layer-fMRI papers out there.

3T layer-fMRI papers that are not cited:

Taso, M., Munsch, F., Zhao, L., Alsop, D.C., 2021. Regional and depth-dependence of cortical blood-flow assessed with high-resolution Arterial Spin Labeling (ASL). *Journal of Cerebral Blood Flow and Metabolism*. <https://doi.org/10.1177/0271678X20982382>

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Strengths:

See above. The authors developed their own SMS sequence with many features. This is important to the field. And does not leave sequence development work to view isolated monopoly labs. This work democratizes SMS.

The questions addressed here are of high relevance to the field: getting tools with good sensitivity, user-friendly applicability, and locally specific brain activity mapping is an important topic in the field of layer-fMRI.

Weaknesses:

(1) I feel the authors need to justify why flow-crushing helps localization specificity. There is an entire family of recent papers that aims to achieve higher localization specificity by doing the exact opposite. Namely, MT or ABC fMRI aims to increase the localization specificity by highlighting the intravascular BOLD by means of suppressing non-flowing tissue. To name a few:

Priovoulos, N., de Oliveira, I.A.F., Poser, B.A., Norris, D.G., van der Zwaag, W., 2023. Combining arterial blood contrast with BOLD increases fMRI intracortical contrast. *Human Brain Mapping* hbm.26227. <https://doi.org/10.1002/hbm.26227>.

Pfaffenrot, V., Koopmans, P.J., 2022. Magnetization Transfer weighted laminar fMRI with multi-echo FLASH. *NeuroImage* 119725. <https://doi.org/10.1016/j.neuroimage.2022.119725>

Schulz, J., Fazal, Z., Metere, R., Marques, J.P., Norris, D.G., 2020. Arterial blood contrast (ABC) enabled by magnetization transfer (MT): a novel MRI technique for enhancing the measurement of brain activation changes. *bioRxiv*. <https://doi.org/10.1101/2020.05.20.106666>

Based on this literature, it seems that the proposed method will make the vein problem worse, not better. The authors could make it clearer how they reason that making GE-BOLD signals more extra-vascular weighted should help to reduce large vein effects.

The empirical evidence for the claim that flow crushing helps with the localization specificity should be made clearer. The response magnitude with and without flow crushing looks pretty much identical to me (see Fig. 6d).

It's unclear to me what to look for in Fig. 5. I cannot discern any layer patterns in these maps. It's too noisy. The two maps of TE=43ms look like identical copies from each other. Maybe an editorial error?

The authors discuss bipolar crushing with respect to SE-BOLD where it has been previously applied. For SE-BOLD at UHF, a substantial portion of the vein signal comes from the intravascular compartment. So I agree that for SE-BOLD, it makes sense to crush the intravascular signal. For GE-BOLD however, this reasoning does not hold. For GE-BOLD (even at 3T), most of the vein signal comes from extravascular dephasing around large unspecific veins and the bipolar crushing is not expected to help with this.

(2) The bipolar crushing is limited to one single direction of flow. This introduces a lot of artificial variance across the cortical folding pattern. This is not mentioned in the manuscript. There is an entire family of papers that perform layer-fMRI with black-blood imaging that solves this with a 3D contrast preparation (VAPER) that is applied across a longer time period, thus killing the blood signal while it flows across all directions of the vascular tree. Here, the signal crushing is happening with a 2D readout as a "snap-shot" crushing. This does not allow the blood to flow in multiple directions.

VAPER also accounts for BOLD contaminations of larger draining veins by means of a tag-control sampling. The proposed approach here does not account for this contamination.

Chai, Y., Li, L., Huber, L., Poser, B.A., Bandettini, P.A., 2020. Integrated VASO and perfusion contrast: A new tool for laminar functional MRI. *NeuroImage* 207, 116358. <https://doi.org/10.1016/j.neuroimage.2019.116358>

Chai, Y., Liu, T.T., Marrett, S., Li, L., Khojandi, A., Handwerker, D.A., Alink, A., Muckli, L., Bandettini, P.A., 2021. Topographical and laminar distribution of audiovisual processing within human planum temporale. *Progress in Neurobiology* 102121. <https://doi.org/10.1016/j.pneurobio.2021.102121>

If I would recommend anyone to perform layer-fMRI with blood crushing, it seems that VAPER is the superior approach. The authors could make it clearer why users might want to use the unidirectional crushing instead.

(3) The comparison with VASO is misleading.

The authors claim that previous VASO approaches were limited by TRs of 8.2s. The authors might be advised to check the latest literature of the last years.

Koiso et al. has performed whole brain layer-fMRI VASO at 0.8mm at 3.9 seconds (with reliable activation) and 2.7 seconds (with unconvincing activation pattern, though), and 2.3 (without activation).

Also, whole brain layer-fMRI BOLD at 0.5mm and 0.7mm has been previously performed by the Juelich group at TRs of 3.5s (their TR definition is 'fishy' though).

Koiso, K., Müller, A.K., Akamatsu, K., Dresbach, S., Gulban, O.F., Goebel, R., Miyawaki, Y., Poser, B.A., Huber, L., 2023. Acquisition and processing methods of whole-brain layer-fMRI VASO and BOLD: The Kenshu dataset. *Aperture Neuro* 34. <https://doi.org/10.1101/2022.08.19.504502>

Yun, S.D., Pais-Roldán, P., Palomero-Gallagher, N., Shah, N.J., 2022. Mapping of whole-cerebrum resting-state networks using ultra-high resolution acquisition protocols. *Human Brain Mapping*. <https://doi.org/10.1002/hbm.25855>

Pais-Roldan, P., Yun, S.D., Palomero-Gallagher, N., Shah, N.J., 2023. Cortical depth-dependent human fMRI of resting-state networks using EPIK. *Front. Neurosci.* 17, 1151544. <https://doi.org/10.3389/fnins.2023.1151544>

The authors are correct that VASO is not advised as a turn-key method for lower brain areas, incl. Hippocampus and subcortex. However, the authors use this word of caution that is intended for inexperienced "users" as a statement that this cannot be performed. This statement is taken out of context. This statement is not from the academic literature. It's advice for the 40+ user base that want to perform layer-fMRI as a plug-and-play routine tool in neuroscience usage. In fact, sub-millimeter VASO is routinely being performed by MRI-physicists across all brain areas (including deep brain structures, hippocampus etc). E.g. see Koiso et al. and an overview lecture from a layer-fMRI workshop that I had recently attended: <https://youtu.be/kzh-nWXd54s?si=hoIjLLIxFUJ4g20&t=2401>

Thus, the authors could embed this phrasing into the context of their own method that they are proposing in the manuscript. E.g. the authors could state whether they think that their sequence has the potential to be disseminated across sites, considering that it requires slow offline reconstruction in Matlab?

Do the authors think that the results shown in Fig. 6c are suggesting turn-key acquisition of a routine mapping tool? In my humble opinion it looks like random noise, with most of the activation outside the ROI (in white matter).

(4) The repeatability of the results is questionable.

The authors perform experiments about the robustness of the method (line 620). The corresponding results are not suggesting any robustness to me. In fact the layer profiles in Fig. 4c vs. Fig 4d are completely opposite. Location of peaks turn into locations of dips and vice versa.

The methods are not described in enough detail to reproduce these results.

The authors mention that their image reconstruction is done "using in-house MATLAB code" (line 634). They do not post a link to github, nor do they say if they share this code.

It is not trivial to get good phase data for fMRI. The authors do not mention how they perform the respective coil-combination.

No data are shared for reproduction of the analysis.

(5) The application of NODRIC is not validated.

Previous applications of NORDIC at 3T layer-fMRI have resulted in mixed success. When not adjusted for the right SNR regime it can result in artifactual reductions of beta scores, depending on the SNR across layers. The authors could validate their application of NORDIC and confirm that the average layer-profiles are unaffected by the application of NORDIC. Also, the NORDIC version should be explicitly mentioned in the manuscript.

Akbari, A., Gati, J.S., Zeman, P., Liem, B., Menon, R.S., 2023. Layer Dependence of Monocular and Binocular Responses in Human Ocular Dominance Columns at 7T using VASO and BOLD (preprint). *Neuroscience*. <https://doi.org/10.1101/2023.04.06.535924>

Knudsen, L., Guo, F., Huang, J., Blicher, J.U., Lund, T.E., Zhou, Y., Zhang, P., Yang, Y., 2023. The laminar pattern of proprioceptive activation in human primary motor cortex. *bioRxiv*. <https://doi.org/10.1101/2023.10.29.564658>

Comments on revisions:

Among all the concerns mentioned above, I think there is only one of the specific issues that was sufficiently addressed.

The authors implemented a combination of three consecutive-dimensional flow crushers. Other concerns were not sufficiently addressed to change my confidence level of the study.

- While the abstract is still focusing on the utility of using 3T, they do not give credit to early 3T layer-fMRI papers leading the way to larger coverage and connectivity applications.

- While the author's choice of using custom SMS 2D readout is justified for them. I do not think that this very method will utilize widespread 3T whole brain connectivity experiments across the global 3T community. This lowers the impact of the paper.

- The images in Fig. 5 are still suspiciously similar. To the level that the noise pattern outside the brain is identical across large parts of the maps with and without PR.

- Maybe it's my ignorance, but I still do not agree why flow crushing focuses the local BOLD responses to small vessels.

- While my feel of a misleading representation of the literature had been accompanied by explicit references, the authors claim that they cannot find them?!? Or claim that they are about something else (which they are not, in my viewpoint).

Data and software are still not shared (not even example data, or nii data).

<https://doi.org/10.7554/eLife.92805.2.sa3>

Reviewer #2 (Public review):

This study developed a setup for laminar fMRI at 3T that aimed to get the best from all worlds in terms of brain coverage, temporal resolution, sensitivity to detect functional responses and spatial specificity. They used a gradient-echo EPI readout to facilitate sensitivity, brain coverage and temporal resolution. The former was additionally boosted by NORDIC denoising and the latter two were further supported by acceleration both in-plane and across slices. The authors evaluated whether the implementation of velocity-nulling (VN) gradients could mitigate macrovascular bias, known to hamper laminar specificity of gradient-echo BOLD.

Strengths:

The setup includes 0.9 mm isotropic acquisitions with large coverage at a reasonable TR. These parameters are hard to optimize simultaneously, and I applaud the ambitious attempt to get "the best from all worlds" (large coverage, high spatio/temporal resolution, spatial specificity, sensitivity), which is sought after in the field. Also, in terms of the availability of the method, it is favorable that it benefits from lower field strength (additional time for VN-gradient implementation, afforded by longer gray matter T2*). Furthermore, I like that the authors took steps to improve the original manuscript by e.g., collecting more data, adjusting the VN implementation to include flow-suppression along three rather than a single dimension, and adjusting the ROI-definition procedure to avoid circularity issues.

That being said, I still find the evidence weak in terms of this sequence achieving high spatial specificity and sensitivity. The results feel oversold and further validation is needed to make a case for the authors' conclusion that "[...] the potential impact of this development is expected to be extensive across various domains of neuroscience research". This is elaborated in the comments below:

The authors acknowledge that the VN setup in its current form probably does not suppress the impact of most ascending veins (these are also not targeted by phase regression, as most are probably too small to produce sufficiently large phase responses). This seems to limit the theoretical support for the author's claim of reduced inter-layer blurring (e.g. the claim that deep and superficial signals are less coupled with VN gradients than without based on Fig 6-7). This limitation withstanding, the method may still be helpful for limiting laminar dependencies by suppressing pial vein responses (which may carry signal from distant regions and layers that blur into superficial layers if left unsuppressed). Unfortunately, the empirical support of VN gradients suppressing superficial bias seems quite weak and is hard to evaluate. For example, the profiles in Figure 4 does not consistently show clearly less superficial bias when VN gradients are on - this might partly be due to the fact that clear bias was not always present in the profiles even without VN. I suspect this is largely explained by the selection of very small and quite unrepresentative ROIs. The corresponding activation maps appear strongly weighted towards CSF which is not always captured in the profile. I recommend sampling a much larger patch of cortex to more accurately capture the actual underlying bias. In this way, all non-VN profiles should have clear bias which should be clearly suppressed for VN if the method is effective. The authors do evaluate the effect of VN/phase regression based on a large activated region in visual cortex (Fig 5) - why not show laminar profiles from here, which is an obvious way to show the effect on superficial bias? I think such evaluations would be a more direct way of evaluating the methods impact on specificity, and are necessary for subsequent FC evaluations to be convincing.

The phase regression results are described inconsistently. In the results section, the authors, in my opinion, "correctly" acknowledge that phase regression seemed to have a very minor impact. However, in the discussion section it is described as if phase regression was effective in suppressing macrovascular responses (L 553-558), which the results do not support (especially based on profiles in Fig 4). There is barely any difference with/without phase regression, which may be due to the fact that ordinary least squares regression was chosen over a deming model which accounts for noise on the phase regressor. Although the authors correctly mentioned in their "answers to reviewers" that the required noise-ratio between magnitude and phase data can be hard to estimate, attempts of that has been described in previous phase regression studies which showed much larger effects (see e.g. Stanley et al. 2020, Knudsen et al. 2023).

I like that the authors put in additional efforts to provide analyses to validate their NORDIC implementation. However, this needs to be done on the VN setup directly, not the "regular BOLD setup" with $b=0$, since the ability of NORDIC to distinguish signal and noise components depends on CNR which is expected to deviate for these setups. Also, it seems z-scores and confidence intervals were computed based on GLM residuals which may lead to inflated z-values and overly narrow CI's due to reduced degrees of freedom following denoising. The denoised z-maps from Fig 3 indeed look somewhat strange, i.e. seemingly increased false positives (more salt/pepper and a bunch of white matter activation) with very weak hand knob activation. Also, something must be wrong with the CIs on the laminar profiles - they seem extremely narrow despite noise levels obviously being high for highly accelerated 3T submillimeter results extracted from a very small ROI. The authors may consider computing these statistics from variance across trials instead.

Given that the idea of the setup is to take advantage in terms of sensitivity by using GE-BOLD contrast relative to e.g. SE-EPI or CBV-weighted setups, they need to carefully demonstrate the sensitivity of their setup, which could be limited by high acceleration factors, the VN gradients, low field strength, etc. I like that they now put more emphasis on non-masked activation maps, but further comparison could be made through tSNR maps, raw single-volume images, raw timeseries, CNR based on across-trial variance, etc.

The major rationale for the setup is to achieve functional connectivity (FC) with brain-wide coverage at laminar resolutions, but it is framed as if this is something that has not been possible in the past with existing setups (statements such as: "Despite advancements in acquisition speed, current CBV/CBF-based fMRI techniques remain inadequate for layer-dependent resting-state fMRI" (L138-140). To me, the functional connectivity results presented here with the VN setup are clearly less convincing than what has been shown with e.g. CBV-weighted acquisitions (e.g. Huber et al. 2021, Chai et al. 2024). The VN setup might also have advantages such as larger coverage as mentioned by the authors, but they fail to balance the comparison by highlighting where previous studies had clear edges. Thus, the impact of the results needs to be down-stated and a more balanced comparison with existing laminar FC studies is warranted. For example, acknowledging that the CBV-weighted studies demonstrate much higher spatial specificity.

Overall I would recommend a stronger emphasis on validating the claims about the sequence on task-based data for which there is a large body of literature to benchmark against (e.g. laminar fMRI studies in V1 and M1), before going to FC where the base for comparison and reference is much more limited in humans at laminar scales.

<https://doi.org/10.7554/eLife.92805.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors are looking for a spatially specific functional brain response to visualise non-invasively with 3T (clinical field strength) MRI. They propose a velocity-nulled weighting to remove signal from draining veins in a submillimeter multiband acquisition.

Strengths:

- This manuscript addresses a real need in the cognitive neuroscience community interested in imaging responses in cortical layers in-vivo in humans.
- An additional benefit is the proposed implementation at 3T, a widely available field strength.

Weaknesses:

- The comparison in Figure 4 for different b-values shows % signal changes. However, as the baseline signal changes with added diffusion weighting, this is rather uninformative. A plot of t-values against cortical depth would be more insightful.
- Surprisingly, the %-signal change for a b-value of 0 is below 1% for 3/4 participants, even at the cortical surface. This raises some doubts about the task or ROI definition. A finger-tapping task should reliably engage the primary motor cortex, even at 3T, and even in individual participants.
- The double peak pattern in the BOLD weighted images in Figure 4 is unexpected given the existing literature on BOLD responses as a function of cortical depth.
- Although I'd like to applaud the authors for their ambition with the connectivity analysis, the low significance threshold used in these maps ($z=1,64$) leads to concerns about the SNR of the underlying data.

I remain unconvinced of the conclusion that the developed VN fMRI exhibited layer specificity - the double peak which is taken as a marker of specificity is not absent in the BOLD responses either, and overall BOLD and VN response profiles as a function of cortical depth are quite similar.

Author response:

The following is the authors' response to the original reviews.

General responses:

The authors sincerely thank all the reviewers for their valuable and constructive comments. We also apologize for the long delay in providing this rebuttal due to logistical and funding challenges. In this revision, we modified the bipolar gradients from one single direction to all three directions. Additionally, in response to the concerns regarding data reliability, we conducted a thorough examination of each step in our data processing pipeline. In the original processing workflow, the projection-onto-convex-set (POCS) method was used for partial Fourier reconstruction. Upon examination, we found that applying the POCS method after parallel image reconstruction significantly altered the signal and resulted in considerable loss of functional feature. Furthermore, the original scan protocol employed a TE of 46 ms, which is notably longer than the typical TE of 33 ms. A prolonged TE can increase the ratio of extravascular to intravascular contributions. Importantly, the impact of TE on the efficacy of phase regression remains unclear, introducing potential confounding effects. To address these issues, we revised the protocol by shortening the TE from 46 ms to 39 ms. This adjustment was achieved by modifying the SMS factor to 3 and the in-plane acceleration rate to 3, thereby minimizing the confounding effects associated with an extended TE.

Following these changes, we recollected task-based fMRI data (N=4) and resting-state fMRI data (N=14) under the updated protocol. Using the revised dataset, we validated layer-specific functional connectivity (FC) through seed-based analyses. These analyses revealed distinct connectivity patterns in the superficial and deep layers of the primary motor cortex (M1), with statistically significant inter-layer differences. Furthermore, additional analyses with a seed in the primary sensory cortex (S1) corroborated the robustness and reliability of the revised methodology. We also changed the 'directed' functional connectivity in the title to 'layer-specific' functional connectivity, as drawing conclusions about directionality requires auxiliary evidence beyond the scope of this study.

We provide detailed responses to the reviewers' comments below.

Reviewer #1 (Public Review):

Summary:

(1) This study aims to provide imaging methods for users of the field of human layer-fMRI. This is an emerging field with 240 papers published so far. Different than implied in the manuscript, 3T is well represented among those papers. E.g. see the papers below that are not cited in the manuscript. Thus, the claim on the impact of developing 3T methodology for wider dissemination is not justified. Specifically, because some of the previous papers perform whole brain layer-fMRI (also at 3T) in more efficient, and more established procedures.

3T layer-fMRI papers that are not cited:

Taso, M., Munsch, F., Zhao, L., Alsop, D.C., 2021. Regional and depth-dependence of cortical blood-flow assessed with high-resolution Arterial Spin Labeling (ASL). *Journal of Cerebral Blood Flow and Metabolism*. <https://doi.org/10.1177/0271678X20982382>

Wu, P.Y., Chu, Y.H., Lin, J.F.L., Kuo, W.J., Lin, F.H., 2018. Feature-dependent intrinsic functional connectivity across cortical depths in the human auditory cortex. *Scientific*

Reports 8, 1-14. <https://doi.org/10.1038/s41598-018-31292-x>

Lifshits, S., Tomer, O., Shamir, I., Barazany, D., Tsarfaty, G., Rosset, S., Assaf, Y., 2018. Resolution considerations in imaging of the cortical layers. *NeuroImage* 164, 112-120. <https://doi.org/10.1016/j.neuroimage.2017.02.086>

Puckett, A.M., Aquino, K.M., Robinson, P.A., Breakspear, M., Schira, M.M., 2016. The spatiotemporal hemodynamic response function for depth-dependent functional imaging of human cortex. *NeuroImage* 139, 240-248. <https://doi.org/10.1016/j.neuroimage.2016.06.019>

Olman, C.A., Inati, S., Heeger, D.J., 2007. The effect of large veins on spatial localization with GE BOLD at 3 T: Displacement, not blurring. *NeuroImage* 34, 1126-1135. <https://doi.org/10.1016/j.neuroimage.2006.08.045>

Ress, D., Glover, G.H., Liu, J., Wandell, B., 2007. Laminar profiles of functional activity in the human brain. *NeuroImage* 34, 74-84. <https://doi.org/10.1016/j.neuroimage.2006.08.020>

Huber, L., Kronbichler, L., Stirnberg, R., Ehse, P., Stocker, T., Fernández-Cabello, S., Poser, B.A., Kronbichler, M., 2023. Evaluating the capabilities and challenges of layer-fMRI VASO at 3T. *Aperture Neuro* 3. <https://doi.org/10.52294/001c.85117>

Scheeringa, R., Bonnefond, M., van Mourik, T., Jensen, O., Norris, D.G., Koopmans, P.J., 2022. Relating neural oscillations to laminar fMRI connectivity in visual cortex. *Cerebral Cortex*. <https://doi.org/10.1093/cercor/bhac154>

We thank the reviewer for listing out 8 papers related to 3T layer-fMRI papers. The primary goal of our work is to develop a methodology for brain-wide, layer-dependent resting-state functional connectivity at 3T. Upon review of the cited papers, we found that:

- (1) One study (Lifshits et al.) was not an fMRI study.
- (2) One study (Olman et al.) was conducted at 7T, not 3T.
- (3) Two studies (Taso et al. and Wu et al.) employed relatively large voxel sizes ($1.6 \times 2.3 \times 5$ mm³ and 1.5 mm isotropic, respectively), which limits layer specificity.
- (4) Only one of the listed studies (Huber et al., Aperture Neuro 2023) provides coverage of more than half of the brain.

While each of these studies offers valuable insights, the VASO study by Huber et al. is the most relevant to our work, given its brain-wide coverage. However, the VASO method employs a relatively long TR (14.137 s), which may not be optimal for resting-state functional connectivity analyses.

To address these limitations, our proposed method achieves submillimeter resolution, layer specificity, brain-wide coverage, and a significantly shorter TR (<5 s) altogether. We believe this advancement provides a meaningful contribution to the field, enabling broader applicability of layer-fMRI at 3T.

(2) The authors implemented a sequence with lots of nice features. Including their own SMS EPI, diffusion bipolar pulses, eye-saturation bands, and they built their own reconstruction around it. This is not trivial. Only a few labs around the world have this level of engineering expertise. I applaud this technical achievement. However, I doubt that any of this is the right tool for layer-fMRI, nor does it represent an advancement for the field. In the thermal noise dominated regime of sub-millimeter fMRI (especially at 3T), it is established to use 3D readouts over 2D (SMS) readouts. While it is not trivial to

implement SMS, the vendor implementations (as well as the CMRR and MGH implementations) are most widely applied across the majority of current fMRI studies already. The author's work on this does not serve any previous shortcomings in the field.

We would like to thank the reviewer for their comments and the recognition of the technical efforts in implementing our sequence. We would like to address the points raised:

(1) We completely agree that in-house implementation of existing techniques does not constitute an advancement for the field. We did not claim otherwise in the manuscript. Our focus was on the development of a method for brain-wide, layer-dependent resting-state functional connectivity at 3T, as mentioned in the response above.

(2) The reviewer stated that "it is established to use 3D readouts over 2D (SMS) readouts". This is a strong claim, and we believe it requires robust evidence to support it. While it is true that 3D readouts can achieve higher tSNR in certain regions, such as the central brain, as shown in the study by Vizioli et al. (ISMRM 2020 abstract; https://cds.ismrm.org/protected/20MProceedings/PDFfiles/3825.html?utm_source=chatgpt.com), higher tSNR does not necessarily equate to improved detection power in fMRI studies. For instance, Le Ster et al. (PLOS ONE, 2019; <https://doi.org/10.1371/journal.pone.0225286>) demonstrated that while 3D EPI had higher tSNR in the central brain, SMS EPI produced higher t-scores in activation maps.

(3) When choosing between SMS EPI and 3D EPI, multiple factors should be taken into account, not just tSNR. For example, SMS EPI and 3D EPI differ in their sensitivity to motion and the complexity of motion correction. The choice between them depends on the specific research goals and practical constraints.

(4) We are open to different readout strategies, provided they can be demonstrated suitable to the research goals. In this study, we opted for 2D SMS primarily due to logistical considerations. This choice does not preclude the potential use of 3D readouts in the future if they are deemed more appropriate for the project objectives.

The mechanism to use bi-polar gradients to increase the localization specificity is doubtful to me. In my understanding, killing the intra-vascular BOLD should make it less specific. Also, the empirical data do not suggest a higher localization specificity to me.

We will elaborate the mechanism and reasoning in the later responses.

Embedding this work in the literature of previous methods is incomplete. Recent trends of vessel signal manipulation with ABC or VAPER are not mentioned. Comparisons with VASO are outdated and incorrect.

The reproducibility of the methods and the result is doubtful (see below).

In this revision, we updated the scan protocol and recollected the imaging data. Detailed explanations and revised results are provided in the later responses.

I don't think that this manuscript is in the top 50% of the 240 layer-fMRI papers out there.

We respect the reviewer's personal opinion. However, we can only address scientific comments or critiques.

Strengths:

See above. The authors developed their own SMS sequence with many features. This is important to the field. And does not leave sequence development work to view isolated

monopoly labs. This work democratizes SMS.

The questions addressed here are of high relevance to the field: getting tools with good sensitivity, user-friendly applicability, and locally specific brain activity mapping is an important topic in the field of layer-fMRI.

Weaknesses:

(1) I feel the authors need to justify why flow-crushing helps localization specificity. There is an entire family of recent papers that aim to achieve higher localization specificity by doing the exact opposite. Namely, MT or ABC fMRI aims to increase the localization specificity by highlighting the intravascular BOLD by means of suppressing non-flowing tissue. To name a few:

Priovoulos, N., de Oliveira, I.A.F., Poser, B.A., Norris, D.G., van der Zwaag, W., 2023. Combining arterial blood contrast with BOLD increases fMRI intracortical contrast. Human Brain Mapping hbm.26227. <https://doi.org/10.1002/hbm.26227>.

Pfaffenrot, V., Koopmans, P.J., 2022. Magnetization Transfer weighted laminar fMRI with multi-echo FLASH. NeuroImage 119725. <https://doi.org/10.1016/j.neuroimage.2022.119725>

Schulz, J., Fazal, Z., Metere, R., Marques, J.P., Norris, D.G., 2020. Arterial blood contrast (ABC) enabled by magnetization transfer (MT): a novel MRI technique for enhancing the measurement of brain activation changes. bioRxiv. <https://doi.org/10.1101/2020.05.20.106666>

Based on this literature, it seems that the proposed method will make the vein problem worse, not better. The authors could make it clearer how they reason that making GE-BOLD signals more extra-vascular weighted should help to reduce large vein effects.

The proposed VN fMRI method employs VN gradients to selectively suppress signals from fast-flowing blood in large vessels. Although this approach may initially appear to diverge from the principles of CBV-based techniques (Chai et al., 2020; Huber et al., 2017a; Pfaffenrot and Koopmans, 2022; Priovoulos et al., 2023), which enhance sensitivity to vascular changes in arterioles, capillaries, and venules while attenuating signals from static tissue and large veins, it aligns with the fundamental objective of all layer-specific fMRI methods. Specifically, these approaches aim to maximize spatial specificity by preserving signals proximal to neural activation sites and minimizing contributions from distal sources, irrespective of whether the signals are intra- or extra-vascular in origin. In the context of intravascular signals, CBV-based methods preferentially enhance sensitivity to functional changes in small vessels (proximal components) while demonstrating reduced sensitivity to functional changes in large vessels (distal components). For extravascular signals, functional changes are a mixture of proximal and distal influences. While tissue oxygenation near neural activation sites represents a proximal contribution, extravascular signal contamination from large pial veins reflects distal effects that are spatially remote from the site of neuronal activity. CBV-based techniques mitigate this challenge by unselectively suppressing signals from static tissues, thereby highlighting contributions from small vessels. In contrast, the VN fMRI method employs a targeted suppression strategy, selectively attenuating signals from large vessels (distal components) while preserving those from small vessels (proximal components). Furthermore, the use of a 3T scanner and the inclusion of phase regression in the VN approach mitigates contamination from large pial veins (distal components) while preserving signals reflecting local tissue oxygenation (proximal components). By integrating these mechanisms, VN fMRI improves spatial specificity, minimizing both intravascular and extravascular contributions that are distal to neuronal activation sites. We have incorporated the responses into Discussion section.

The empirical evidence for the claim that flow crushing helps with the localization specificity should be made clearer. The response magnitude with and without flow crushing looks pretty much identical to me (see Fig. 6d).

In the new results in Figure 4, the application of VN gradients attenuated the bias towards pial surface. Consistent with the results in Figure 4, Figure 5 also demonstrated the suppression of macrovascular signal by VN gradients.

It's unclear to me what to look for in Fig. 5. I cannot discern any layer patterns in these maps. It's too noisy. The two maps of TE=43ms look like identical copies from each other. Maybe an editorial error?

In this revision, the original Figure 5 has been removed. However, we would like to clarify that the two maps with TE = 43 ms in the original Figure 5 were not identical. This can be observed in the difference map provided in the right panel of the figure.

The authors discuss bipolar crushing with respect to SE-BOLD where it has been previously applied. For SE-BOLD at UHF, a substantial portion of the vein signal comes from the intravascular compartment. So I agree that for SE-BOLD, it makes sense to crush the intravascular signal. For GE-BOLD however, this reasoning does not hold. For GE-BOLD (even at 3T), most of the vein signal comes from extravascular dephasing around large unspecific veins, and the bipolar crushing is not expected to help with this.

The reviewer's statement that "most of the vein signal comes from extravascular dephasing around large unspecific veins" may hold true for 7T. However, at 3T, the susceptibility-induced Larmor frequency shift is reduced by 57%, and the extravascular contribution decreases by more than 35%, as shown by Uludağ et al. 2009 (DOI: 10.1016/j.neuroimage.2009.05.051).

Additionally, according to the biophysical models (Ogawa et al., 1993; doi: 10.1016/S0006-3495(93)81441-3), the extravascular contamination from the pial surface is inversely proportional to the square of the distance from vessel. For a vessel diameter of 0.3 mm and an isotropic voxel size of 0.9 mm, the induced frequency shift is reduced by at least 36-fold at the next voxel. Notably, a vessel diameter of 0.3 mm is larger than most pial vessels. Theoretically, the extravascular effect contributes minimally to inter-layer dependency, particularly at 3T compared to 7T due to weaker susceptibility-related effects at lower field strengths. Empirically, as shown in Figure 7c, the results at M1 demonstrated that layer specificity can be achieved statistically with the application of VN gradients. We have incorporated this explanation into the Introduction and Discussion sections of the manuscript.

(2) The bipolar crushing is limited to one single direction of flow. This introduces a lot of artificial variance across the cortical folding pattern. This is not mentioned in the manuscript. There is an entire family of papers that perform layer-fMRI with black-blood imaging that solves this with a 3D contrast preparation (VAPER) that is applied across a longer time period, thus killing the blood signal while it flows across all directions of the vascular tree. Here, the signal crushing is happening with a 2D readout as a "snap-shot" crushing. This does not allow the blood to flow in multiple directions.

VAPER also accounts for BOLD contaminations of larger draining veins by means of a tag-control sampling. The proposed approach here does not account for this contamination.

Chai, Y., Li, L., Huber, L., Poser, B.A., Bandettini, P.A., 2020. Integrated VASO and perfusion contrast: A new tool for laminar functional MRI. *NeuroImage* 207, 116358. <https://doi.org/10.1016/j.neuroimage.2019.116358>

Chai, Y., Liu, T.T., Marrett, S., Li, L., Khojandi, A., Handwerker, D.A., Alink, A., Muckli, L., Bandettini, P.A., 2021. Topographical and laminar distribution of audiovisual processing within human planum temporale. *Progress in Neurobiology* 102121. <https://doi.org/10.1016/j.pneurobio.2021.102121>

If I would recommend anyone to perform layer-fMRI with blood crushing, it seems that VAPER is the superior approach. The authors could make it clearer why users might want to use the unidirectional crushing instead.

We understand the reviewer's concern regarding the directional limitation of bipolar crushing. As noted in the responses above, we have updated the bipolar gradient to include three orthogonal directions instead of a single direction. Furthermore, flow-related signal suppression does not necessarily require a longer time period. Bipolar diffusion gradients have been effectively used to nullify signals from fast-flowing blood, as demonstrated by Boxerman et al. (1995; DOI: 10.1002/mrm.1910340103). Their study showed that vessels with flow velocities producing phase changes greater than π radians due to bipolar gradients experience significant signal attenuation. The critical velocity for such attenuation can be calculated using the formula: $1/(2gGd)$ where g is the gyromagnetic ratio, G is the gradient strength, d is the gradient pulse width and D is the time between the two bipolar gradient pulses. In the framework of Boxerman et al. at 1.5T, the critical velocity for b value of 10 s/mm^2 is ~ 8 mm/s, resulting in a $\sim 30\%$ reduction in functional signal. In our 3T study, b values of 6, 7, and 8 s/mm^2 correspond to critical velocities of 16.8, 15.2, and 13.9 mm/s, respectively. The flow velocities in capillaries and most venules remain well below these thresholds. Notably, in our VN fMRI sequences, bipolar gradients were applied in all three orthogonal directions, whereas in Boxerman et al.'s study, the gradients were applied only in the z -direction. Given the voxel dimensions of $3 \times 3 \times 7 \text{ mm}^3$ in the 1.5T study, vessels within a large voxel are likely oriented in multiple directions, meaning that only a subset of fast-flowing signals would be attenuated. Therefore, our approach is expected to induce greater signal reduction, even at the same b values as those used in Boxerman et al.'s study. We have incorporated this text into the Discussion section of the manuscript.

(3) The comparison with VASO is misleading.

The authors claim that previous VASO approaches were limited by TRs of 8.2s. The authors might be advised to check the latest literature of the last years.

Koiso et al. performed whole brain layer-fMRI VASO at 0.8mm at 3.9 seconds (with reliable activation), 2.7 seconds (with unconvincing activation pattern, though), and 2.3 (without activation).

Also, whole brain layer-fMRI BOLD at 0.5mm and 0.7mm has been previously performed by the Juelich group at TRs of 3.5s (their TR definition is 'fishy' though).

Koiso, K., Müller, A.K., Akamatsu, K., Dresbach, S., Gulban, O.F., Goebel, R., Miyawaki, Y., Poser, B.A., Huber, L., 2023. Acquisition and processing methods of whole-brain layer-fMRI VASO and BOLD: The Kenshu dataset. *Aperture Neuro* 34. <https://doi.org/10.1101/2022.08.19.504502>

Yun, S.D., Pais-Roldán, P., Palomero-Gallagher, N., Shah, N.J., 2022. Mapping of whole-cerebrum resting-state networks using ultra-high resolution acquisition protocols. *Human Brain Mapping*. <https://doi.org/10.1002/hbm.25855>

Pais-Roldan, P., Yun, S.D., Palomero-Gallagher, N., Shah, N.J., 2023. Cortical depth-dependent human fMRI of resting-state networks using EPIK. *Front. Neurosci.* 17, 1151544. <https://doi.org/10.3389/fnins.2023.1151544>

We thank the reviewer for providing these references. While the protocol with a TR of 3.9 seconds in Koiso's work demonstrated reasonable activation patterns, it was not tested for layer specificity. Given that higher acceleration factors (AF) can cause spatial blurring, a protocol should only be eligible for comparison if layer specificity is demonstrated.

Secondly, the TRs reported in Koiso's study pertain only to either the VASO or BOLD acquisition, not the combined CBV-based contrast. To generate CBV-based images, both VASO and BOLD data are required, effectively doubling the TR. For instance, if the protocol with a TR of 3.9 seconds is used, the effective TR becomes approximately 8 seconds. The stable protocol used by Koiso et al. to acquire whole-brain data (94.08 mm along the z-axis) required 5.2 seconds for VASO and 5.1 seconds for BOLD, resulting in an effective TR of 10.3 seconds. The spatial resolution achieved was 0.84 mm isotropic.

Unfortunately, we could not find the Juelich paper mentioned by the reviewer.

To have a more comprehensive comparison, we collated relevant literature on brain-wide layer-specific fMRI. We defined brain-wide acquisition as imaging protocols that cover more than half of the human brain, specifically exceeding 55 mm along the superior-inferior axis. We identified five studies and summarized their scan parameters, including effective TR, coverage, and spatial resolution, in Table 1.

The authors are correct that VASO is not advised as a turn-key method for lower brain areas, incl. Hippocampus and subcortex. However, the authors use this word of caution that is intended for inexperienced "users" as a statement that this cannot be performed. This statement is taken out of context. This statement is not from the academic literature. It's advice for the 40+ user base that wants to perform layer-fMRI as a plug-and-play routine tool in neuroscience usage. In fact, sub-millimeter VASO is routinely being performed by MRI-physicists across all brain areas (including deep brain structures, hippocampus etc). E.g. see Koiso et al. and an overview lecture from a layer-fMRI workshop that I had recently attended: <https://youtu.be/kzh-nWXd54s?si=hoIjJLLIxFUJ4g20&t=2401>

In this revision, we decided to focus on cortico-cortical functional connectivity and have removed the LGN-related content. Consequently, the text mentioned by the reviewer was also removed. Nevertheless, we apologize if our original description gave the impression that functional mapping of deep brain regions using VASO is not feasible. The word of caution we used is based on the layer-fMRI blog (https://layerfmri.com/2021/02/22/vaso_ve/) and reflects the challenges associated with this technique, as outlined by experts like Dr. Huber and Dr. Strinberg.

According to the information provided, including the video, functional mapping of the hippocampus and amygdala using VASO is indeed possible but remains technically challenging. The short arterial arrival times in these deep brain regions can complicate the acquisition, requiring RF inversion pulses to cover a wider area at the base of the brain. For example, as of 2023, four or more research groups were attempting to implement layer-fMRI VASO in the hippocampus. One such study at 3T required multiple inversion times to account for inflow effects, highlighting the technical complexity of these applications. This is the context in which we used the word of caution. We are not sure whether recent advancements like MAGEC VASO have improved its applicability. As of 2024, we have not identified any published VASO studies specifically targeting deep brain structures such as the hippocampus

or amygdala. Therefore, it is difficult to conclude that “sub-millimeter VASO is routinely being performed by MRI physicists on deep brain structures such as the hippocampus.”

Thus, the authors could embed this phrasing into the context of their own method that they are proposing in the manuscript. E.g. the authors could state whether they think that their sequence has the potential to be disseminated across sites, considering that it requires slow offline reconstruction in Matlab?

We are enthusiastic about sharing our imaging sequence, provided its usefulness is conclusively established. However, it's important to note that without an online reconstruction capability, such as the ICE, the practical utility of the sequence may be limited. Unfortunately, we currently don't have the manpower to implement the online reconstruction. Nevertheless, we are more than willing to share the offline reconstruction codes upon request.

Do the authors think that the results shown in Fig. 6c are suggesting turn-key acquisition of a routine mapping tool? In my humble opinion, it looks like random noise, with most of the activation outside the ROI (in white matter).

As we mentioned in the ‘general response’ in the beginning of the rebuttal, the POCS method for partial Fourier reconstruction caused the loss of functional feature, potentially accounting for the activation in white matter. In this revision, we have modified the pulse sequence, scan protocol and processing pipelines.

According to the results in Figure 4, stable activation in M1 was observed at the single-subject level across most scan protocols. Yet, the layer-dependent activation profiles in M1 were spatially unstable, irrespective of the application of VN gradients. This spatial instability is not entirely unexpected, as T2*-based contrast is inherently sensitive to various factors that perturb the magnetic field, such as eye movements, respiration, and macrovascular signal fluctuations. Furthermore, ICA-based artifact removal was intentionally omitted in Figure 4 to ensure fair comparisons between protocols, leaving residual artifacts unaddressed. Inconsistency in performing the button-pressing task across sessions may also have contributed to the observed variability. These results suggest that submillimeter-resolution fMRI may not yet be suitable for reliable individual-level layer-dependent functional mapping, unless group-level statistics are incorporated to enhance robustness. We have incorporated this text into the Limitation section of the manuscript.

(4) The repeatability of the results is questionable.

The authors perform experiments about the robustness of the method (line 620). The corresponding results are not suggesting any robustness to me. In fact, the layer profiles in Fig. 4c vs. Fig 4d are completely opposite. The location of peaks turns into locations of dips and vice versa.

The methods are not described in enough detail to reproduce these results.

The authors mention that their image reconstruction is done "using in-house MATLAB code" (line 634). They do not post a link to github, nor do they say if they share this code.

We thank the reviewer for the comments regarding reproducibility and data sharing. In response, we have revised the Methods section and elaborated on the technical details to improve clarity and reproducibility.

Regarding code sharing, we acknowledge that the current in-house MATLAB reconstruction code requires further refinement to improve its readability and usability. Due to limited manpower, we have not yet been able to complete this task. However, we are committed to

making the code publicly available and will upload it to GitHub as soon as the necessary resources are available.

For data sharing, we face logistical challenges due to the large size of the dataset, which spans tens of terabytes. Platforms like OpenNeuro, for example, typically support datasets up to 10TB, making it difficult to share the data in its entirety. Despite this limitation, we are more than willing to share offline reconstruction codes and raw data upon request to facilitate reproducibility.

Regarding data robustness, we kindly refer the reviewer to our response to the previous comment, where we addressed these concerns in greater detail.

It is not trivial to get good phase data for fMRI. The authors do not mention how they perform the respective coil-combination.

No data are shared for reproduction of the analysis.

Obtaining phase data is relatively straightforward when the images are retrieved directly from raw data. For coil combination, we employed the adaptive coil combination approach described by (Walsh et al.; DOI: 10.1002/(sici)1522-2594(200005)43:5<682::aid-mrm10>3.0.co;2-g) The MATLAB code for this implementation was developed by Dr. Diego Hernando and is publicly available at <https://github.com/welton0411/matlab>.

(5) The application of NORDIC is not validated.

Previous applications of NORDIC at 3T layer-fMRI have resulted in mixed success. When not adjusted for the right SNR regime it can result in artifactual reductions of beta scores, depending on the SNR across layers. The authors could validate their application of NORDIC and confirm that the average layer-profiles are unaffected by the application of NORDIC. Also, the NORDIC version should be explicitly mentioned in the manuscript.

Akbari, A., Gati, J.S., Zeman, P., Liem, B., Menon, R.S., 2023. Layer Dependence of Monocular and Binocular Responses in Human Ocular Dominance Columns at 7T using VASO and BOLD (preprint). *Neuroscience*. <https://doi.org/10.1101/2023.04.06.535924>

Knudsen, L., Guo, F., Huang, J., Blicher, J.U., Lund, T.E., Zhou, Y., Zhang, P., Yang, Y., 2023. The laminar pattern of proprioceptive activation in human primary motor cortex. *bioRxiv*. <https://doi.org/10.1101/2023.10.29.564658>

We appreciate the reviewer's suggestion. To validate the application of NORDIC denoising in our study, we compared the BOLD activation maps before and after denoising in the visual and motor cortices, as well as the depth-dependent activation profiles in M1. These results are presented in Figure 3. The activation patterns in the denoised maps were consistent with those in the non-denoised maps but exhibited higher statistical significance. Notably, BOLD activation within M1 was only observed after NORDIC denoising, underscoring the necessity of this approach. Figure 3c shows the depth-dependent activation profiles in M1, highlighted by the green contours in Figure 3b. Both denoised and non-denoised profiles followed similar trends; however, as expected, the non-denoised profile exhibited larger confidence intervals compared to the NORDIC-denoised profile. These results confirm that NORDIC denoising enhances sensitivity without introducing distortions in the functional signal. The corresponding text has been incorporated into the Results section.

Regarding the implementation details of NORDIC denoising, the reconstructed images were denoised using a g-factor map (function name: NIFTI_NORDIC). The g-factor map was estimated from the image time series, and the input images were complex-valued. The width of the smoothing filter for the phase was set to 10, while all other hyperparameters were

retained at their default values. This information has been integrated into the Methods section for clarity and reproducibility.

Reviewer #2 (Public Review):

This study developed a setup for laminar fMRI at 3T that aimed to get the best from all worlds in terms of brain coverage, temporal resolution, sensitivity to detect functional responses, and spatial specificity. They used a gradient-echo EPI readout to facilitate sensitivity, brain coverage and temporal resolution. The former was additionally boosted by NORDIC denoising and the latter two were further supported by parallel-imaging acceleration both in-plane and across slices. The authors evaluated whether the implementation of velocity-nulling (VN) gradients could mitigate macrovascular bias, known to hamper the laminar specificity of gradient-echo BOLD.

The setup allows for 0.9 mm isotropic acquisitions with large coverage at a reasonable TR (at least for block designs) and the fMRI results presented here were acquired within practical scan-times of 12-18 minutes. Also, in terms of the availability of the method, it is favorable that it benefits from lower field strength (additional time for VN-gradient implementation, afforded by longer gray matter T2).*

The well-known double peak feature in M1 during finger tapping was used as a test-bed to evaluate the spatial specificity. They were indeed able to demonstrate two distinct peaks in group-level laminar profiles extracted from M1 during finger tapping, which was largely free from superficial bias. This is rather intriguing as, even at 7T, clear peaks are usually only seen with spatially specific non-BOLD sequences. This is in line with their simple simulations, which nicely illustrated that, in theory, intravascular macrovascular signals should be suppressible with only minimal suppression of microvasculature when small b-values of the VN gradients are employed. However, the authors do not state how ROIs were defined making the validity of this finding unclear; were they defined from independent criteria or were they selected based on the region mostly expressing the double peak, which would clearly be circular? In any case, results are based on a very small sub-region of M1 in a single slice - it would be useful to see the generalizability of superficial-bias-free BOLD responses across a larger portion of M1.

We appreciate and understand the reviewer's concerns. Given the small size of the hand knob region within M1 and its intersubject variability in location, defining this region automatically remains challenging. However, we applied specific criteria to minimize bias during the delineation of M1: 1) the hand knob region was required to be anatomically located in the precentral sulcus or gyrus; 2) it needed to exhibit consistent BOLD activation across the majority of testing conditions; and 3) the region was expected to show BOLD activation in the deep cortical layers under the condition of $b = 0$ and $TE = 30$ ms. Once the boundaries across cortical depth were defined, the gray matter boundaries of hand knob region were delineated based on the T1-weighted anatomical image and the cortical ribbon mask but excluded the BOLD activation map to minimize potential bias in manual delineation. Based on the new criteria, the resulting depth-dependent profiles, as shown in Figure 4, are no longer superficial-bias-free.

As repeatedly mentioned by the authors, a laminar fMRI setup must demonstrate adequate functional sensitivity to detect (in this case) BOLD responses. The sensitivity evaluation is unfortunately quite weak. It is mainly based on the argument that significant activation was found in a challenging sub-cortical region (LGN). However, it was a single participant, the activation map was not very convincing, and the demonstration of significant activation after considerable voxel-averaging is inadequate evidence to claim sufficient BOLD sensitivity. How well sensitivity is retained in the presence of VN gradients, high acceleration factors, etc., is therefore unclear. The ability

of the setup to obtain meaningful functional connectivity results is reassuring, yet, more elaborate comparison with e.g., the conventional BOLD setup (no VN gradients) is warranted, for example by comparison of tSNR, quantification and comparison of CNR, illustration of unmasked-full-slice activation maps to compare noise-levels, comparison of the across-trial variance in each subject, etc. Furthermore, as NORDIC appears to be a cornerstone to enable submillimeter resolution in this setup at 3T, it is critical to evaluate its impact on the data through comparison with non-denoised data, which is currently lacking.

We appreciate the reviewer's comments and acknowledge that the LGN results from a single participant were not sufficiently convincing. In this revision, we have removed the LGN-related results and focused on cortico-cortical FC. To evaluate data quality, we opted to present BOLD activation maps rather than tSNR, as high tSNR does not necessarily translate to high functional significance. In Figure 3, we illustrate the effect of NORDIC denoising, including activation maps and depth-dependent profiles. Figure 4 presents activation maps acquired under different TE and b values, demonstrating that VN gradients effectively reduce the bias toward the pial surface without altering the overall activation patterns. The results in Figure 4 and Figure 5 provide evidence that VN gradients retain sensitivity while reducing superficial bias. The ability of the setup to obtain meaningful FC results was validated through seed-based analyses, identifying distinct connectivity patterns in the superficial and deep layers of the primary motor cortex (M1), with significant inter-layer differences (see Figure 7). Further analyses with a seed in the primary sensory cortex (S1) demonstrated the reliability of the method (see Figure 8). For further details on the results, including the impact of VN gradients and NORDIC denoising, please refer to Figures 3 to 8 in the Results section.

Additionally, we acknowledge the limitations of our current protocol for submillimeter-resolution fMRI at the individual level. We found that robust layer-dependent functional mapping often requires group-level statistics to enhance reliability. This issue has been discussed in detail in the Limitations section.

The proposed setup might potentially be valuable to the field, which is continuously searching for techniques to achieve laminar specificity in gradient echo EPI acquisitions. Nonetheless, the above considerations need to be tackled to make a convincing case.

Reviewer #3 (Public Review):

Summary:

The authors are looking for a spatially specific functional brain response to visualise non-invasively with 3T (clinical field strength) MRI. They propose a velocity-nulled weighting to remove the signal from draining veins in a submillimeter multiband acquisition.

Strengths:

- This manuscript addresses a real need in the cognitive neuroscience community interested in imaging responses in cortical layers in-vivo in humans.*
- An additional benefit is the proposed implementation at 3T, a widely available field strength.*

Weaknesses:

- Although the VASO acquisition is discussed in the introduction section, the VN-sequence seems closer to diffusion-weighted functional MRI. The authors should make it more clear to the reader what the differences are, and how results are expected to differ. Generally, it is not so clear why the introduction is so focused on the VASO acquisition (which, curiously, lacks a reference to Lu et al 2013). There are many more alternatives to*

BOLD-weighted imaging for fMRI. CBF-weighted ASL and GRASE have been around for a while, ABC and double-SE have been proposed more recently.

The major distinction between diffusion-weighted fMRI (DW-fMRI) and our methodology lies in the b-value employed. DW-fMRI typically measures cellular swelling using b-values greater than 1000 s/mm² (e.g., 1800 s/mm²). In contrast, our VN-fMRI approach measures hemodynamic responses by employing smaller b-values specifically designed to suppress signals from fast-flowing draining veins rather than detecting microstructural changes.

Regarding other functional contrasts, we agree that more layer-dependent fMRI approaches should be mentioned. In this revision, we have expanded the Introduction section to include discussions of the double spin-echo approach and CBV-based methods, such as MT-weighted fMRI, VAPER, ABC, and CBF-based method ASL. Additionally, the reference to Lu et al. (2013) has been cited in the revised manuscript. The corresponding text has been incorporated into the Introduction section to provide a more comprehensive overview of alternative functional imaging techniques.

- *The comparison in Figure 2 for different b-values shows % signal changes. However, as the baseline signal changes dramatically with added diffusion weighting, this is rather uninformative. A plot of t-values against cortical depth would be much more insightful.*
- *Surprisingly, the %-signal change for a b-value of 0 is not significantly different from 0 in the gray matter. This raises some doubts about the task or ROI definition. A finger-tapping task should reliably engage the primary motor cortex, even at 3T, and even in a single participant.*
- *The BOLD weighted images in Figure 3 show a very clear double-peak pattern. This contradicts the results in Figure 2 and is unexpected given the existing literature on BOLD responses as a function of cortical depth.*
- *Given that data from Figures 2, 3, and 4 are derived from a single participant each, order and attention affects might have dramatically affected the observed patterns. Especially for Figure 4, neither BOLD nor VN profiles are really different from 0, and without statistical values or inter-subject averaging, these cannot be used to draw conclusions from.*

We appreciate the reviewer's suggestions. In this revision, we have made significant updates to the participant recruitment, scan protocol, data processing, and M1 delineation. Please refer to the "General Responses" at the beginning of the rebuttal and the first response to Reviewer #2 for more details.

Previously, the variation in depth-dependent profiles was calculated across upscaled voxels within a specific layer. However, due to the small size of the hand knob region, the number of within-layer voxels was limited, resulting in inaccurate estimations of signal variation. In the revised manuscript, the signal was averaged within each layer before performing the GLM analysis, and signal variation was calculated using the temporal residuals. The technical details of these changes are described in the "Materials and Methods" section. Furthermore, while the initial submission used percentage signal change for the profiles of M1, the dramatic baseline fluctuations observed previously are no longer an issue after the modifications. For this reason, we retained the use of percentage signal change to present the depth-dependent profiles. After these adjustments, the profiles exhibited a bias toward the pial surface, particularly in the absence of VN gradients.

- *In Figure 5, a phase regression is added to the data presented in Figure 4. However, for a phase regression to work, there has to be a (macrovascular) response to start with. As none of the responses in Figure 4 are significant for the single participant dataset, phase*

regression should probably not have been undertaken. In this case, the functional 'responses' appear to increase with phase regression, which is contra-intuitive and deserves an explanation.

We agreed with reviewer's argument. In the revised results, the issues mentioned by the reviewer are largely diminished. The updated analyses demonstrate that phase regression effectively reduces superficial bias, as shown in Figures 4 and 5.

- Consistency of responses is indeed expected to increase by a removal of the more variable vascular component. However, the microvascular component is always expected to be smaller than the combination of microvascular + macrovascular responses. Note that the use of %signal changes may obscure this effect somewhat because of the modified baseline. Another expected feature of BOLD profiles containing both micro- and microvasculature is the draining towards the cortical surface. In the profiles shown in Figure 7, this is completely absent. In the group data, no significant responses to the task are shown anywhere in the cortical ribbon.

We agreed with reviewer's comments. In the revised manuscript, the results have been substantially updated to addressing the concerns raised. The original Figure 7 is no longer relevant and has been removed.

- Although I'd like to applaud the authors for their ambition with the connectivity analysis, I feel that acquisitions that are so SNR starved as to fail to show a significant response to a motor task should not be used for brain wide directed connectivity analysis.

We appreciate the reviewer's comments and share the concern about SNR limitations. In the updated results presented in Figure 5, the activation patterns in the visual cortex were consistent across TEs and b values. At the motor cortex, stable activation in M1 was observed at the single-subject level across most scan protocols. However, the layer-dependent activation profiles in M1 exhibited spatial instability, irrespective of the application of VN gradients. This spatial instability is not entirely unexpected, as T2*-based contrast is inherently sensitive to factors that perturb the magnetic field, such as eye movements, respiration, and macrovascular signal fluctuations. Additionally, ICA-based artifact removal was intentionally omitted in Figure 4 to ensure fair comparisons across protocols, leaving some residual artifacts unaddressed. Variability in task performance during button-pressing sessions may have further contributed to the observed inconsistencies.

Although these findings suggest that submillimeter-resolution fMRI may not yet be reliable for individual-level layer-dependent functional mapping, the group-level FC analyses can still yield robust results. In Figure 7, group-level statistics revealed distinct functional connectivity (FC) patterns associated with superficial and deep layers in M1. These FC maps exhibited significant differences between layers, demonstrating that VN fMRI enhances inter-layer independence. Additional FC analyses with a seed placed in S1 further validated these findings (see Figure 8).

The claim of specificity is supported by the observation of the double-peak pattern in the motor cortex, previously shown in multiple non-BOLD studies. However, this same pattern is shown in some of the BOLD weighted data, which seems to suggest that the double-peak pattern is not solely due to the added velocity nulling gradients. In addition, the well-known draining towards the cortical surface is not replicated for the BOLD-weighted data in Figures 3, 4, or 7. This puts some doubt about the data actually having the SNR to draw conclusions about the observed patterns.

We appreciate the reviewer's comments. In the updated results, the efficacy of the VN gradients is evident near the pial surface, as shown in Figures 4 and 5. In Figure 4, comparing the second and third columns ($b = 0$ and $b = 6 \text{ s/mm}^2$, respectively, at $TE = 38 \text{ ms}$), the percentage signal change in the superficial layers is generally lower with $b = 6 \text{ s/mm}^2$ than with $b = 0$. This indicates that VN gradient-induced signal suppression is more pronounced in the superficial layers. Additionally, in Figure 5, the VN gradients effectively suppressed macrovascular signals as highlighted by the blue circles. These observations support the role of VN gradients in enhancing specificity by reducing superficial bias and macrovascular contamination. Furthermore, bias towards cortical surface was observed in the updated results in Figure 4.

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

(1) L141: "depth dependent" is slightly misleading here. It could be misunderstood to suggest that the authors are assessing how spatial specificity varies as a function of depth. Rather, they are assessing spatial specificity based on depth-dependent responses (double peak feature). Perhaps "layer-dependent spatial specificity" could be substituted with laminar specificity?

We thank the reviewer for the suggestion. The term "depth dependent" has been replaced by "layer dependent" in the revised manuscript.

(2) L146-149: these do not validate spatial specificity.

The original text is removed.

(3) L180: Maybe helpful to describe what the b-value is to assist unfamiliar readers.

We have clarified the b-value as "the strength of the bipolar diffusion gradients" where it is first mentioned in the manuscript.

(4) Figure 1B: I think it would be appropriate with a sentence of how the authors define micro/macrovasculature. Figure 1B seems to suggest that large ascending veins are considered microvascular which I believe is a bit unconventional. Nevertheless, as long as it is clearly stated, it should be fine.

In our context, macrovasculature refers to vessels that are distal to neural activation sites and contribute to extravascular contamination. These vessels are typically larger in size (e.g., $> 0.1 \text{ mm}$ in diameter) and exhibit faster flow rates (e.g., $> 10 \text{ mm/s}$).

(5) I think the authors could be more upfront with the point about non-suppressed extravascular effects from macrovasculature, which was briefly mentioned in the discussion. It could already be highlighted in the introduction or theory section.

We thank the reviewer's suggestions. We have expanded the discussion of extravascular effects from macrovasculature in both the Introduction (5th paragraph) and Discussion (3rd paragraph) sections.

(6) The phase regression figure feels a bit misplaced to me. If the authors agree: rather than showing the TE-dependency of the effect of phase regression, it may be more relevant for the present study to compare the conventional setup with phase regression,

with the VN setup without phase regression. I.e., to show how the proposed setup compares to existing 3T laminar fMRI studies.

In this revision, both the TE-dependent and VN-dependent effects of phase regression were investigated. The results in Figure 4 and Figure 5 demonstrated that phase regression effectively suppresses macrovascular contributions primarily near the gray matter/CSF boundary, irrespective of TE or the presence of VN gradients.

(7) L520: It might be beneficial to also cite the large body of other laminar studies showing the double peak feature to underscore that it is highly robust, which increases its relevance as a test-bed to assess spatial specificity.

We agreed. More literatures have been cited (Chai et al., 2020; Huber et al., 2017a; Knudsen et al., 2023; Priovoulos et al., 2023).

(8) L557: The argument that only one participant was assessed to reduce inter-subject variability is hard to buy. If significant variability exists across subjects, this would be highly relevant to the authors and something they would want to capture.

We thank the reviewer for the suggestions. In this revision, we have increased the number of participants to 4 for protocol development and 14 for resting-state functional connectivity analysis, allowing us to better assess and account for inter-subject variability.

(9) L637: add download link and version number.

The download link has been added as requested. The version number is not applicable.

(10) L638: How was the phase data coil-combined?

The reconstructed multi-channel data, which were of complex values, were combined using the adaptive combination method (Walsh et al.; DOI: 10.1002/(sici)1522-2594(200005)43:5<682::aid-mrm10>3.0.co;2-g). The MATLAB code for this implementation was developed by Dr. Diego Hernando and is publicly available at <https://github.com/welton0411/matlab>. The phase data were then extracted using the MATLAB function 'angle'.

(11) L639: Why was the smoothing filter parameter changed (other parameters were default)?

The smoothing filter parameter was set based on the suggestion provided in the help comments of the NIFTI_NORDIC function:

```
function NIFTI_NORDIC(fn_magn_in,fn_phase_in,fn_out,ARG)

% fMRI

% ARG.phase_filter_width=10;
```

In other words, we simply followed the recommendation outlined in the NIFTI_NORDIC function's documentation.

(12) I assume the phase data was motion corrected after transforming to real and imaginary components and using parameters estimated from magnitude data? Maybe add a few sentences about this.

Prior to phase regression, the time series of real and imaginary components were subjected to motion correction, followed by phase unwrapping. The phase regression was incorporated early in the data processing pipeline to minimize the discrepancy in data processing between magnitude and phase images (Stanley et al., 2021).

(13) Was phase regression applied with e.g., a deming model, which accounts for noise on both the x and y variable? In my experience, this makes a huge difference compared with regular OLS.

We appreciate the reviewer's insightful comment. We are aware that the noise present in both magnitude and phase data therefore linear Deming regression would be a good fit to phase regression (Stanley et al., 2021). To perform Deming regression, however, the ratio of magnitude error variance to phase error variance must be predefined. In our initial tests, we found that the regression results were sensitive to this ratio. To avoid potential confounding, we opted to use OLS regression for the current analysis. However, we agreed Deming model could enhance the efficacy of phase regression if the ratio could be determined objectively and properly.

(14) Figure 2: What is error bar reflecting? I don't think the across-voxel error, as also used in Figure 4, is super meaningful as it assumes the same response of all voxels within a layer (might be alright for such a small ROI). Would it be better to e.g. estimate single-trial response magnitude (percent signal change) and assess variability across? Also, it is not obvious to me why b=30 was chosen. The authors argue that larger values may kill signal, but based on this Figure in isolation, b=48 did not have smaller response magnitudes (larger if anything).

We agreed with the reviewer's opinion on the across-voxel error. In the revised manuscript, the signal was averaged within each layer before performing the GLM analysis, and signal variation was calculated using the temporal residuals. The technical details of these changes are described in the "Materials and Methods" section.

Additionally, the bipolar diffusion gradients were modified from a single direction to three orthogonal directions. As a result, the questions and results related to b=30 or b=48 are no longer applicable.

(15) Figure 5: would be informative to quantify the effect of phase regression over a large ROI and evaluate reduction in macrovascular influence from superficial bias in laminar profiles.

We appreciate the reviewer's suggestion. In the revised manuscript, the reduction in macrovascular influence from superficial bias across a large ROI is displayed in Figure 5. Additionally, the impact on laminar profiles is demonstrated in Figure 4.

(16) L406-408: What kind of robustness?

We acknowledge that describing the protocol as "robust" was an overstatement. The updated results indicate that the current protocol for submillimeter fMRI may not yet be suitable for reliable individual-level layer-dependent functional mapping. However, group-level functional connectivity (FC) analyses demonstrated clear layer-specific distinctions with VN fMRI, which were not evident in conventional fMRI. These findings highlight the enhanced layer specificity achievable with VN fMRI.

(17) Figure 8: I think C) needs pointers to superficial, middle, and deep layers? Why is it not in the same format as in Figure 9C? The discussion of the FC results could benefit

from more references supporting that these observations are in line with the literature.

In the revised results, the layer pooling shown in Figure 9c has been removed, making the question regarding format alignment no longer applicable. Additionally, references supporting the FC results have been added to the revised Discussion section (7th paragraph).

(18) L456-457: But correlation coefficients may also be biased by different CNR across layers.

That is correct. In the updated FC results in Figure 7 to 9, we used group-level statistics rather than correlation coefficients.

Reviewer #3 (Recommendations For The Authors):

The results in Figure 2-6 should be repeated over, or averaged over, a (small) group of participants. N=6 is usual in this field. I would seriously reconsider the multiband acceleration - the acquisition seemingly cannot support the SNR hit.

A few more specific points are given below:

(1) Abstract: The sentence about LGN in the abstract came for me out of the blue - why would LGN be important here, it's not even a motor network node? Perhaps the aims of the study should be made more clear - if it's about networks as suggested earlier then a network analysis result would be expected too. Expanding the directed FC findings would improve the logical flow of the abstract. Given the many concerns, removing the connectivity analysis altogether would also be an option.

We thank the reviewer for the suggestions. The LGN-related results indeed diluted the focus of this study and have been completely removed in this revision.

(2) Line 105: in addition to the VASO method, ..

The corresponding text has been revised, and as a result, the reviewer's suggestion is no longer applicable.

(3) If out of the set MB 4 / 5 / 6 MB4 was best, why did the authors not continue with a comparison including MB3 and MB2? It seems to me unlikely that the MB4 acquisition is actually optimal.

Results: We appreciate the reviewer's suggestions. In this revision, we decreased the MB factor to 3, as it allowed us to increase the in-plane acceleration rate to 3, thereby shortening the TE. The resulting sensitivity for both individual and group-level results is detailed in earlier responses, such as the response to Q16 for Reviewer #2.

(4) The formatting of the references is occasionally flawed, including first names and/or initials. Please consider using a reliable reference manager.

We used Zotero as our reference manager in this revision to ensure consistency and accuracy. The references have been formatted according to the APA style.

(5) In the caption of Figure 5, corrected and uncorrected p values are identical. What multiple comparisons correction was made here? A multiple comparisons over voxels (as is standard) would usually lead to a cut-off $\sim z=3.2$. That would remove most of the 'responses' shown in figure 5.

We appreciate the reviewer's comment. The original results presented in Figure 5 have been removed in the revised manuscript, making this comment no longer applicable.

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